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A 17L LPDI H_2 ICE Concept for the Very Heavy Applications by Volvo Group – Update

Konzept eines 17L LPDI Wasserstoffverbrennungsmotors für schwere Nutzfahrzeuge von Volvo Group

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Abstract

The Volvo group has committed to have a CO_2 neutral product fleet offer by 2040. Important pillars to achieve CO_2 neutrality are the battery electric vehicles and the H_2 fuel cell products, but Volvo also recognises the need to keep combustion engines for certain applications, for example operations with very high up-time and/or in very rough environments with high payloads.

The development of a H_2 engine demonstrator based on the present 780hp 17L diesel platform will be presented in this paper. In the previous paper [1] the digital development was described. Now the focus is to describe the results from testing of the demonstrator engine. Key results include peak torque of 3000 Nm at 1000 rpm, maximum power approaching 600 hp at 1600 rpm, peak brake efficiency exceeding 42% and the possibility of engine -out emissions of NOx at approximate EuroVI tailpipe levels.

Furthermore, the potential of the Volvo Ultra Lean concept will be described and discussed with the latest updates, including potential hybridisation.

Kurzfassung

Die Volvo Gruppe hat sich dazu verpflichtet bis Ende 2040 nur noch CO_2 neutrale Produkte anzubieten. Wichtige Säulen, um diese CO_2 Neutralität zu erreichen sind Batterie und Brennstoffzellenfahrzeuge, darüber hinaus sieht Volvo auch weiterhin den Bedarf von Verbrennungsmotoren für Anwendungen mit hoher Auslastung und/oder extremen Arbeitsbedingungen in Kombination mit hohen Lasten.

Diese Abhandlung beschreibt die Entwicklung eines H_2-Prototypmotors auf Basis der 780PS 17L Dieselmotorplattform. Die vorherige Abhandlung aus dem Jahre 2023 beschrieb die digitale Entwicklung. Der Fokus dieser Abhandlung sind die Testresultate des H_2-Prototypmotors. Zu den wichtigsten Resultaten zählen das maximale Drehmoment von 3000 Nm bei 1000 u/min und die maximale Leistung von 600 PS die bei 1600 u/min erreicht wird. Der maximale Wirkungsgrad überschreitet 42%, trotzdem besteht die Möglichkeit von Stickoxidrohemissionen auf dem Niveau von EuroVI.

Außerdem wird das Potenzial des besonderes mageren (Volvo Ultra Lean) Konzeptes beschrieben, als auch das Potenzial von Hybridisierung.

Introduction

The Volvo group has committed to the Science Based Targets initiative (SBTi), [2] , and therefore has set the goal to become CO_2 neutral by 2040 (Figure 1). Important pillars to achieve this are the battery electric medium and heavy-duty vehicles and heavy-duty H_2 fuel cell products, but Volvo also recognises the need to keep combustion engines for certain applications, for example operations with very high up-time and/or in very rough environments (Figure 2, [3]). Volvo trucks will release H_2 ICE trucks before the end of the decade [4].

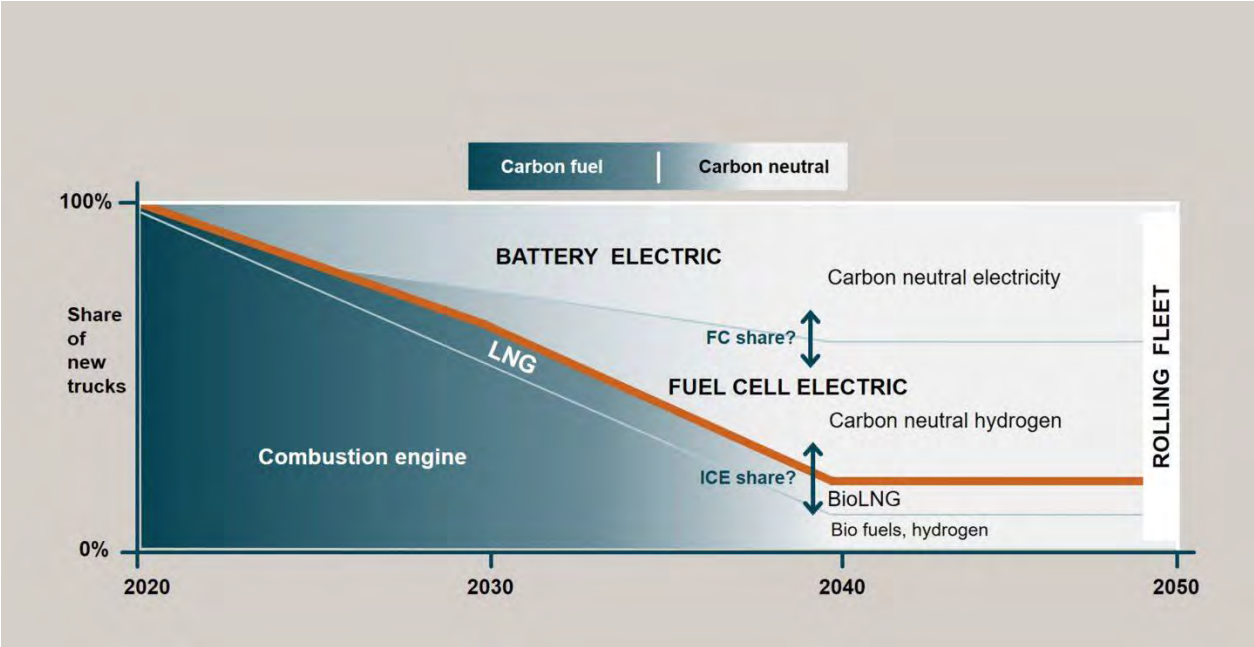

Figure 1: Volvo group, the way to carbon neutrality

Figure 2: A Volvo 16L truck performing a heavy haul operation of a Volvo Wheel loader in a gravel pit

The Volvo Group offers products, including trucks and buses and also off-highway products ranging from boat engines through mining and construction equipment to off-grid power generators. Many of these products are difficult to replace with alternative propulsion like electric drivelines powered by batteries or fuel-cells so the combustion engine will still be needed. Hydrogen is an alternative fuel, that also eliminates CO_2 in the exhaust gas.

Volvo has a long history of evaluating alternative fuels, such as DME [5] and natural gas/LNG. Now there is a recognition of the need for a hydrogen fuelled ICE to complement the diesel/HVO and LNG offerings within traditional powertrains. A parallel track for development of H_2 ICE is Volvo's involvement in the HYCET project [6], which aims to bring hydrogen fuelled long-distance trucks into public road transport service for BMW Logistics by 2025.

There are two main technologies, spark ignited (SI), pre-mixed engines either with port or direct injection of fuel or diffusion combustion with Cespira HPDI technology (High Pressure Direct Injections) [7]. Cespira is a joint venture between Volvo group and Westport fuel systems. The Volvo group today offers LNG products with HPDI technology and is also developing HPDI technology for hydrogen. The first truck product will use HPDI technology [4].

However, recognizing the fact that a small amount of diesel or HVO is used to initiate combustion with HPDI it is also interesting to keep developing CO_2 free technologies.
Therefore, one such development track is the 16L LPDI (Low Pressure Direct Injection) engine as was presented previously in Vienna [1]. In the meantime, the 16L base platform has now been replaced by a 17L diesel engine with up to 780hp [8]. It was therefore natural to base the prototype H_2 engine on the new 17L engine. The prototype engine has been running for most of 2024 in a test-cell within Volvo.

For the H_2 SI engines, port and direct injection has been evaluated, with support from single-cylinder engine experiments. The benefit of higher volumetric efficiency and therefore higher power density directed the choice towards direct injection (LPDI). It does however drive the need for substantial re-design of the cylinder head and requires an injector capable of delivering the hydrogen gas directly into the cylinder while achieving a high level of homogenisation to decrease risk for uncontrolled combustion (knock) and keeping NO_X-formation low.

Because of the high laminar combustion velocity of hydrogen, measures need to be taken to reduce the effect of this. The primary path is to run with very lean mixtures, not only does this decrease combustion velocity and formation of NO_X (lower combustion temperature) but very lean operation also decreases the pumping losses otherwise associated with spark ignition engines (mainly during stoichiometric operation). Setting the performance goals high might require going from merely "lean" mixture to "Ultra Lean" as was shown by Porsche [9]. However, due to the lean operation an exhaust aftertreatment system (EATS) with selective catalytic reduction (SCR) system similar to those on diesel vehicles will be needed.

This paper will focus on the results that were achieved with the demonstrator engine. The engine (Figure 4) was built and run with a traditional dual-stage boosting system (Figure 3), but the potential of a very advanced, possibly electrified, boosting system will also be described. The latter would be capable of Ultra Lean operation including possibility to achieve less NO_X-emissions than the Euro 7 limits of less than 200 mg/kWh [10] during transient operation in world harmonized transient cycle (WHTC).

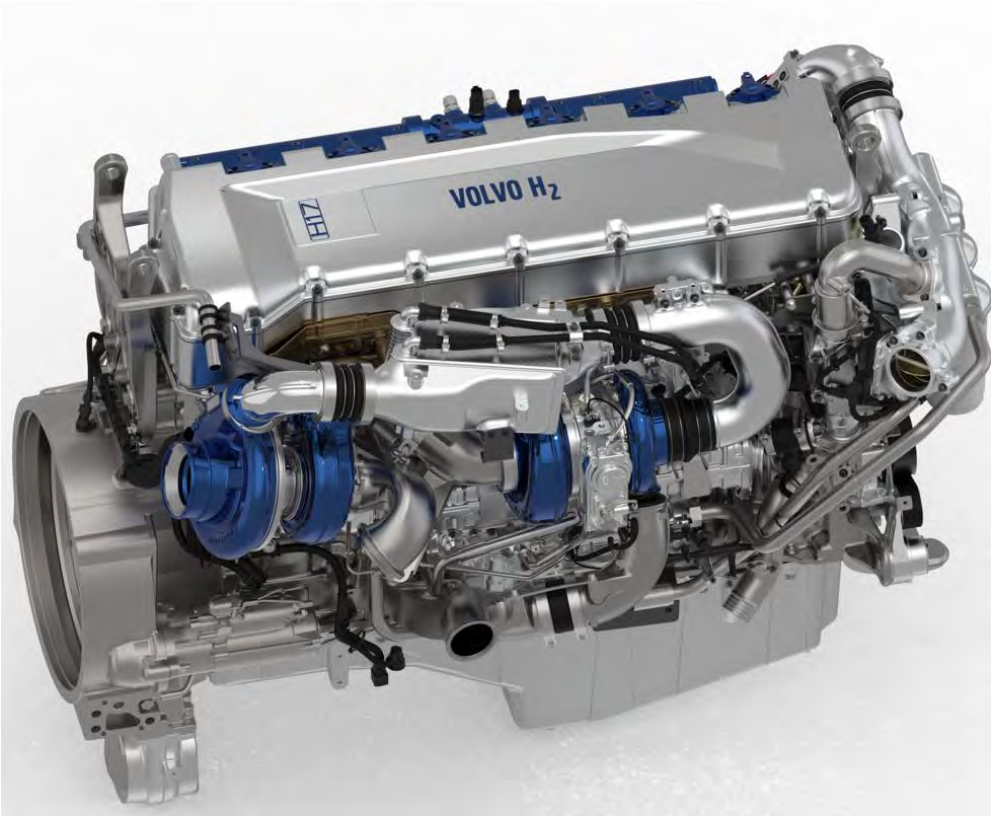

Figure 3: A render of the 17L H_2 engine, showing the dual-stage boosting system

Figure 4: The built 17L H_2 engine at the time of preparation for testing

Engine development

When initiating the development of a hydrogen engine it was logical to evaluate the engine platforms available inside the Volvo Group. With the clear interest to evaluate the ultimate performance potential, it was logical to base this work on the Euro6 StepE,17L diesel engine, recently released with 780 hp in diesel form [8]. From early single-cylinder testing, it became clear that it would be beneficial not only to have the spark plug in the centre but also the injector. This required a substantial re-design of the base cylinder head, involving changes both to the coolant jacket and to the intake and exhaust ports. The resulting design placed both injector and spark plug at slight angles towards the centre of the combustion chamber. All the while keeping the existing valve train, including Volvo Compression Brake (VCB) functionality.

The engine has been in testing during 2024 and below a summary of the results will be presented. The first part will compare the predictions from simulations to the results from actual testing. The second part will present the results more in-depth, both from a combustion perspective and with regards to exhaust emissions. Finally, an update of the potential of Ultra Lean will be given including hardware specifications.

Combustion system

The combustion system design is based on a 16L diesel engine design with tandem style ports, see Figure 5. The packaging situation under the valve hood and in the cylinder-head is difficult for the standard diesel engine. To fit yet another part, the spark plug, is challenging due to the limited available space. In addition, both the positioning of the spark plug and positioning of the injector are vital for the combustion system performance. The positioning of the injector and the design of it does to a large extent govern the mixing process of fuel and air. The spark plug is best situated in the centre of the cylinder to allow for a symmetrical flame propagation. Hence, the initial design hypothesis for the combustion system was that both the spark plug and the injector would benefit from being situated close to the cylinder centre. The valves were adjusted in size and position to allow for centralised positioning of the spark plug and injector. The result as seen from a combustion system point of view is depicted in Figure 6 and Figure 7. As seen in the images neither the spark plug nor the injector is situated in the absolute centre but the positioning is believed to be a good compromise for both parts.

The geometry of the intake ports results in a very low in-cylinder gas motion, with both a tumble and swirl ratio well below 0.5 at the spark timing window around TDC. This gives the conditions at which the fuel-air mixing process will take place, *i.e.* the mixing cannot rely on large scale gas motion but is dependent on a good injector and piston design.

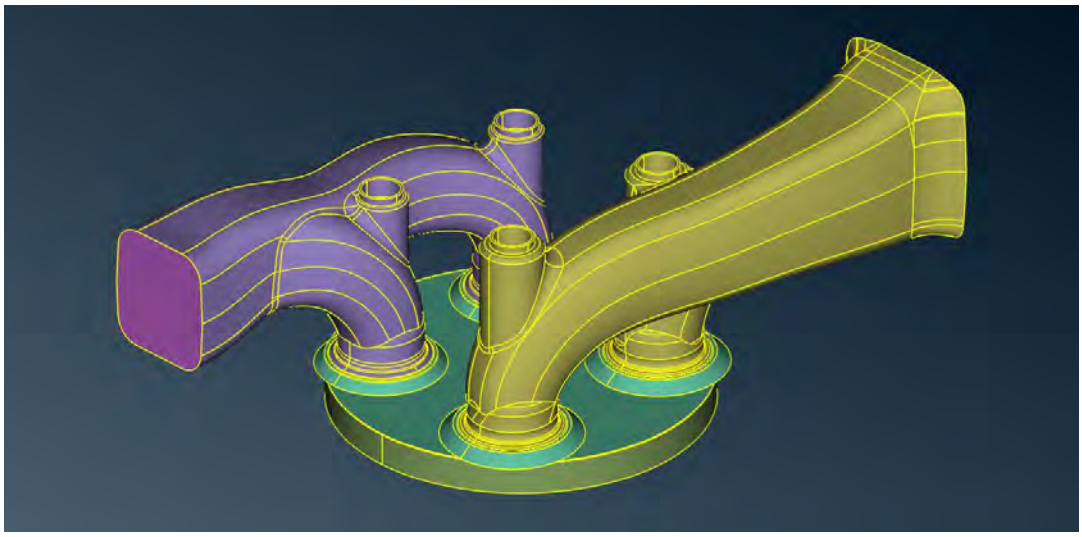

Figure 5: CFD model of the 16L combustion system showing the internal geometry of the intake and exhaust ports.

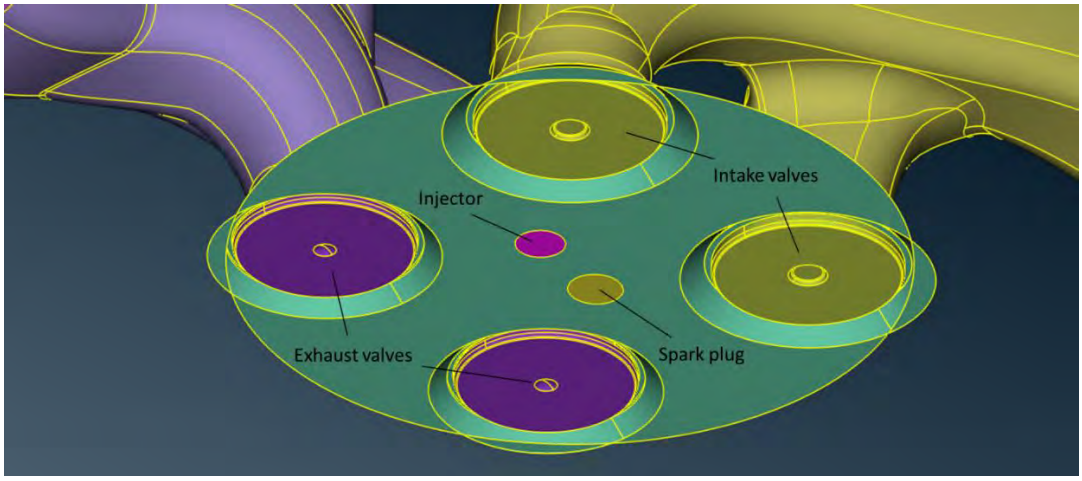

Figure 6: Combustion face of the cylinder head showing valves, injector position and spark plug position.

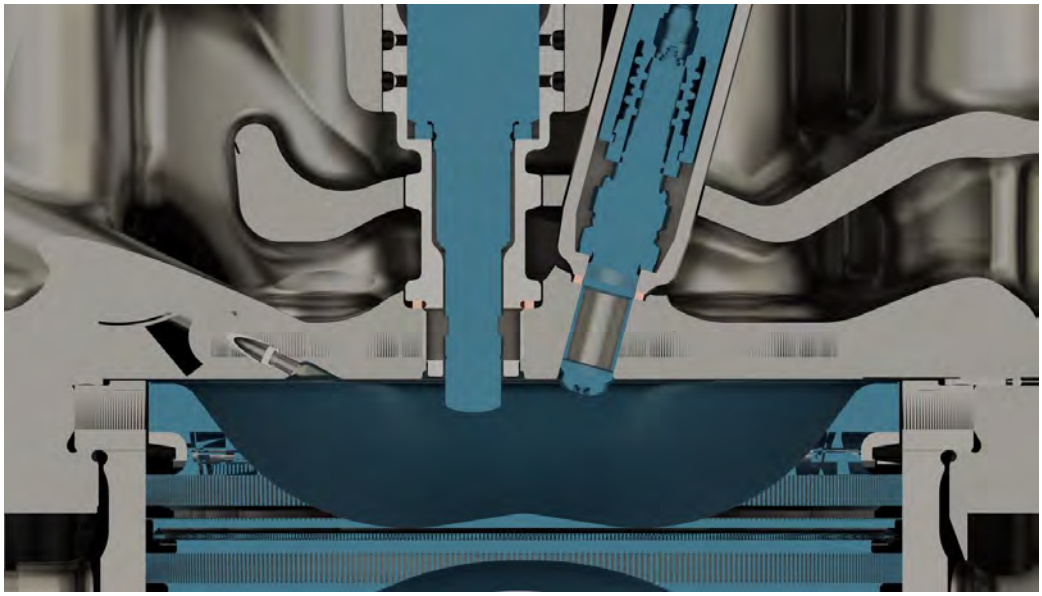

Figure 7: Render of the 17L H_2 cylinder head with injector, spark plug and H_2 specific piston

Simulation – follow up

The previous paper [1] focused on simulation in 1D and 3D. Now, the validity of those simulations can be verified and the models used to tune the precision further. Below is a summary of that work including some key findings.

The 17L engine has been updated with a turbo-system capable of providing diesel-like performance while meeting the H_2 combustion requirements and the capability of running at high lambda values. The performance evaluation performed with engine model in GT-suite presented in the previous paper [1] showed that although with current hardware it should be possible to achieve brake mean effective pressure (BMEP) of 25 bar (760 hp), the targeted power for the current engine is 21 bar BMEP (620 hp). The built engine tested to obtain the targeted torque curve. While testing is still going on, it is very beneficial to re-run the simulation with the set up similar to the tests and compare the results with test. Such comparison is a powerful tool for evaluating the performance and accuracy of the simulation model and tuning and calibrating the model. Therefore, the simulation results will become more predictive and reliable.

Results presented in the previous paper [1] includes the best efficiency points and therefore include best points with exhaust gas recirculation (EGR). However, the tests were run without EGR, therefore some new plots without EGR for the best efficiency points will be presented first also. Results presented in the previous paper [1] includes the best efficiency points and therefore include best points with exhaust gas recirculation (EGR). However, the tests were run without EGR, therefore some new plots without EGR for the best efficiency points will be presented first also. Gross Indicated mean effective pressure (IMEPg) values of the highest measured load points during the engine testing are plotted in Figure 8 in comparison to the highest gross IMEPg calculated from the 1D model with GT-suite.

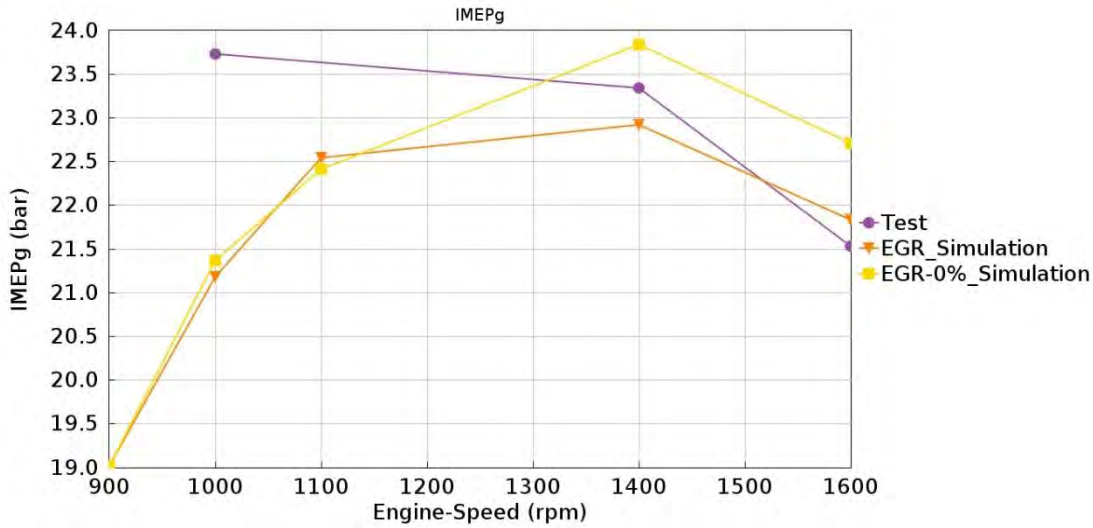

Figure 8: Gross IMEP values (targeted torque curve an test results)

Figure 9 presents the brake efficiency estimation from the model without EGR for all full load points in comparison to the base cases with EGR. Since the brake efficiency varies very little (less than 0.5%) for the various lambda value, the results for highest lambda value are plotted. Figure 9 shows that running the engine at full load without EGR reduced the brake efficiency due to deteriorating the performance of the turbo system.

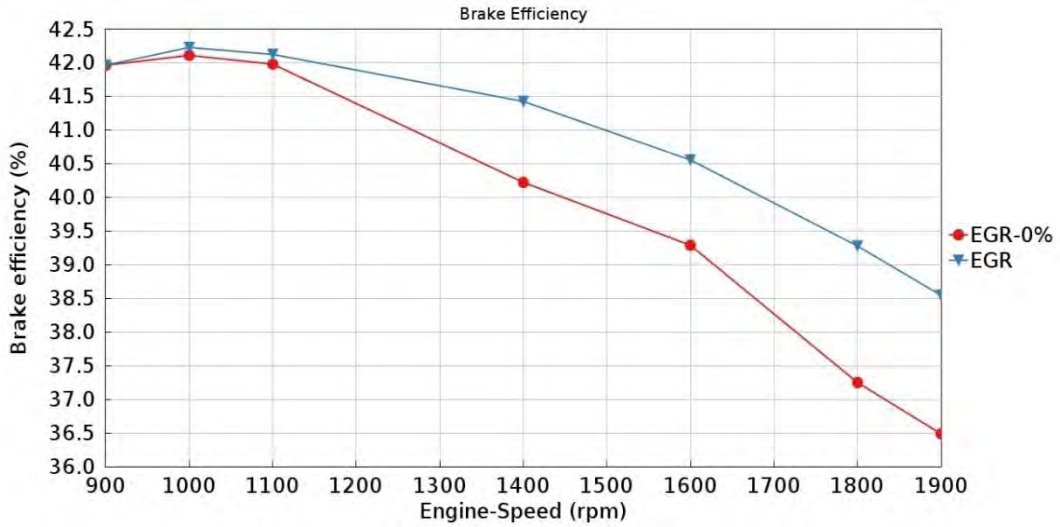

Figure 9: Highest brake efficiency estimation from engine model with and without EGR

In order to compare the capability of the engine model in estimating the engine performance, at first step Three Pressure Analysis (TPA) were performed to tune the actual compression ratio, valve lash values and global heat transfer coefficient. At the next step simulations were done by controlling the model similar to tests. Therefore, test values for injected fuel mass, boost pressure and 50% fuel burned crank angle were targeted in the simulations. The gross IMEP (IMEPg) is not targeted in the simulation, so Figure 10 shows the comparison between the test IMEPg and the Simulated values. The points that are presented in this plot includes the highest power achieved at each engine speed along with the highest lambda value during test.

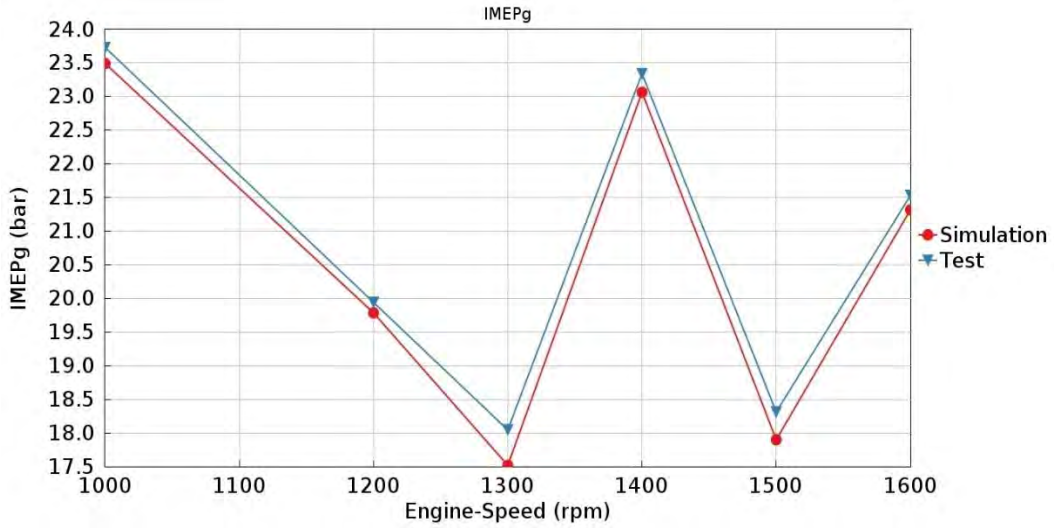

Figure 10: IMEP gross values from test and simulation

Less than 3% of gross IMEP difference is obtained from test and simulation. All load points have lower gross IMEP value in simulation compared to the tests. The reason for obtaining less power from simulation in comparison to test could be explained by the increased value of the exhaust manifold pressure in simulation in comparison to test (Figure 11). Higher exhaust manifold obtained in simulation resulted in higher residual percentage within the cylinder along with higher charge temperature. Therefore, in comparison to test, the fresh air trapped mass and net IMEP (IMEPn) decreased while Pumping mean effective pressure (PMEP) increased as presented in figure Figure 12.

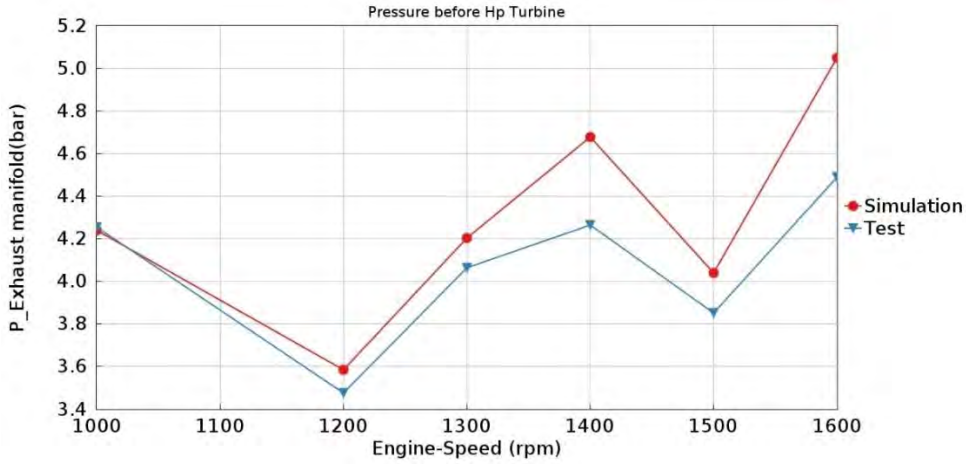

Figure 11: Exhaust manifold pressure from test and simulation

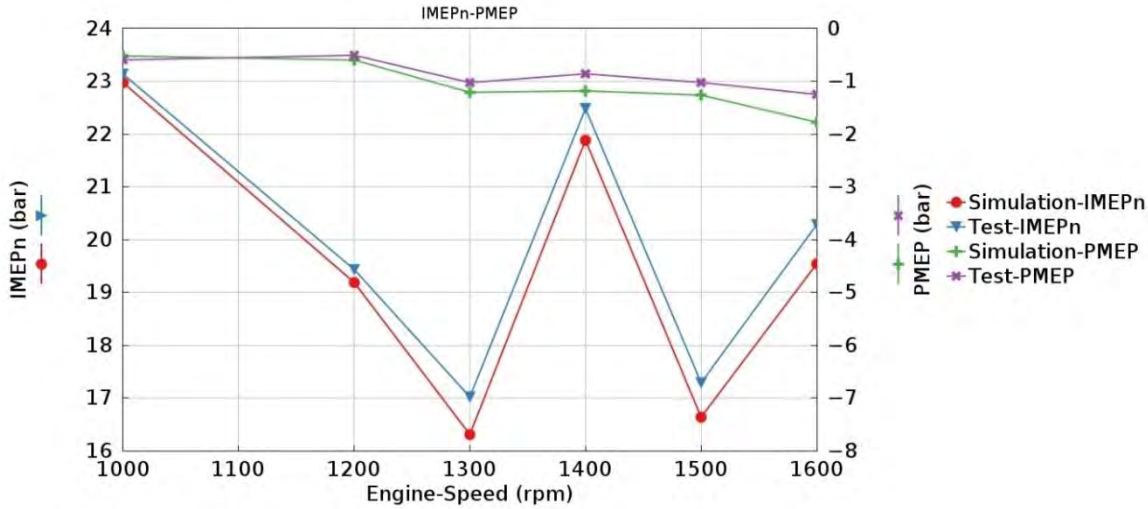

Figure 12: IMEP net and PMEP values from test and simulation

Figure 13 presents the lambda value obtained from simulation in comparison to test. The plotted lambda values are the apparent lambda and were obtained by the ratio of airflow to fuel flow and the effect of residuals are not considered.

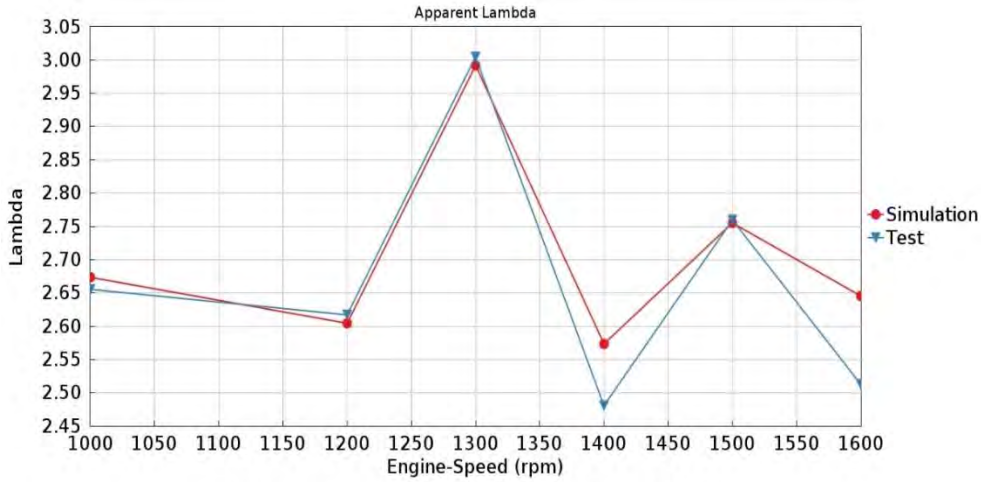

Figure 13: Apparent lambda from test and simulation

For almost all load points same lambda values are obtained from tests and simulations except for 1400 and 1600 rpm. Such difference from test and simulation could be explained by comparing the intake manifold pressure (representing boost pressure) plotted in Figure 14.

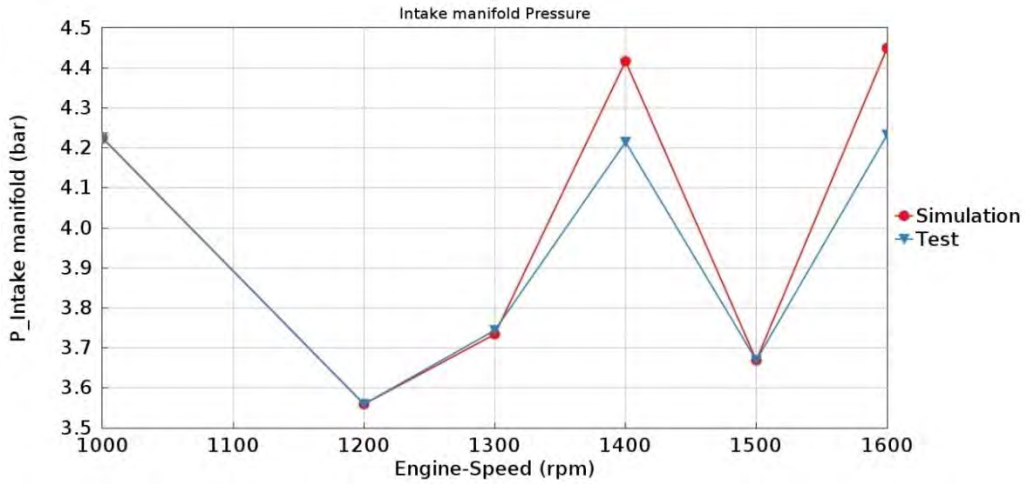

Figure 14: Intake manifold pressure from test and simulation

Figure 14 shows that although the boost pressures were targeted in the simulation, higher boost pressures for load points at 1400 and 1600 rpm are estimated. These two load points have very high power and the rack VGT (Variable geometry turbine) position is fully open. However, still the targeted pressure is not obtained. The reason for such discrepancy arises from different performance of the CAC (Charge air cooler) system in combination with the low-pressure turbo system. Figure 15 shows the difference between the pressure after the low-pressure (LP) compressor between test and simulation.

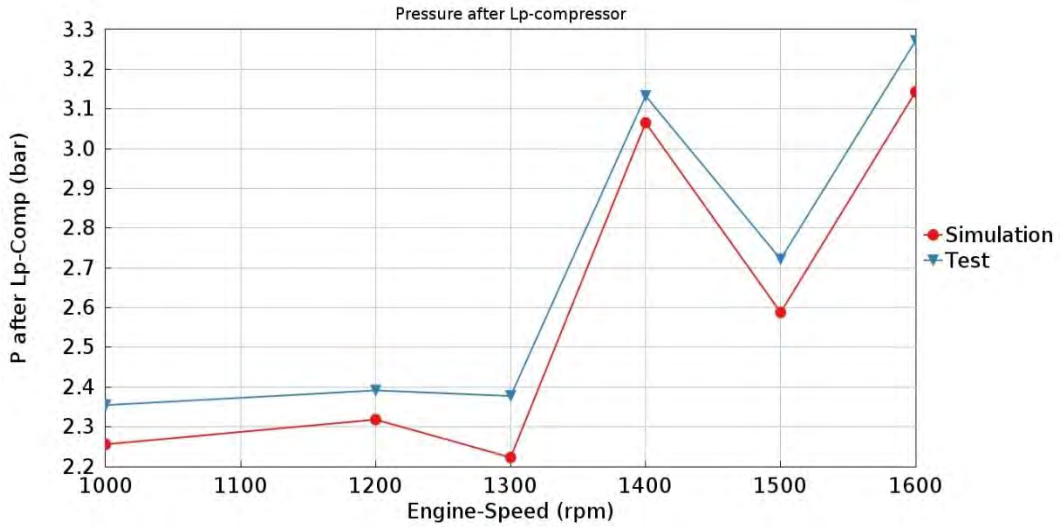

Figure 15: Pressure after the low-pressure compressor from test and simulation

Higher pressure value after low pressure compressor is obtained in test in comparison to simulation. On the other hand, at very high load the air flow is high, however the CAC model is not calibrated for such high flow values. Therefore, the flow condition cannot be fully predicted after the CAC system. Boost pressure is targeted for controlling and adjusting of the rack VGT position. Therefore, high-pressure turbo compensated for boost pressure by closing and adjusting the rack VGT position and to obtain boost pressure similar to test. However, since the rack VGT position is fully open in load points of 1400 and 1600 rpm, lower pressures cannot be obtained by opening the rack VGT. Therefore, at these two load

points, lower boost pressure and higher lambda values are achieved. Figure 16 shows pressure after high pressure (HP) compressor that is adjusted through the VGT system for all load points expect 1400 and 1600 rpm load points.

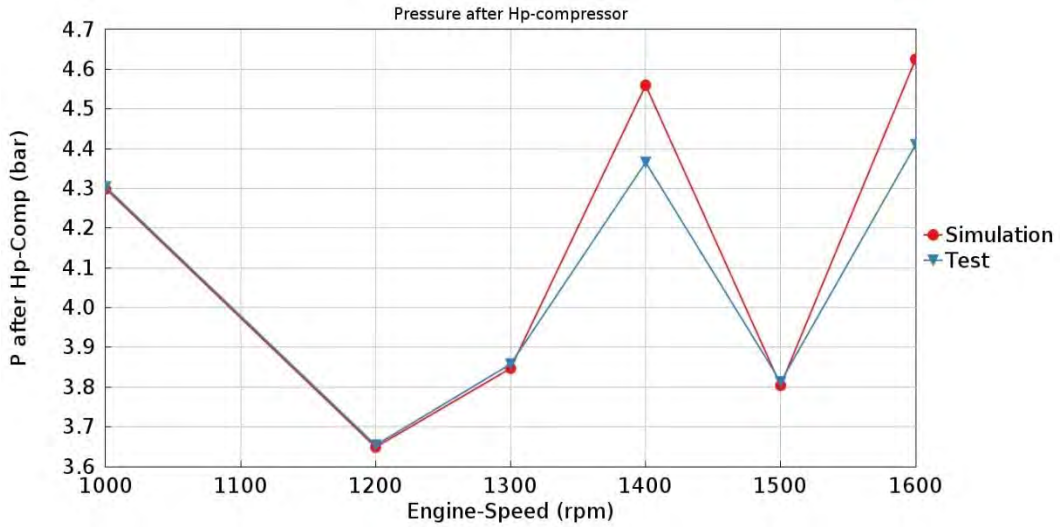

Figure 16: Pressure after the high-pressure compressor from test and simulation

Figure 17 shows the brake efficiency of the test in comparison to simulation. The overall heat transfer coefficient has a considerable effect on the obtained value from simulation. This coefficient is considered to have constant value for all engine speed in simulation however in reality it depends on the engine speed and to calibrate it based on engine speed would provides even better estimation.

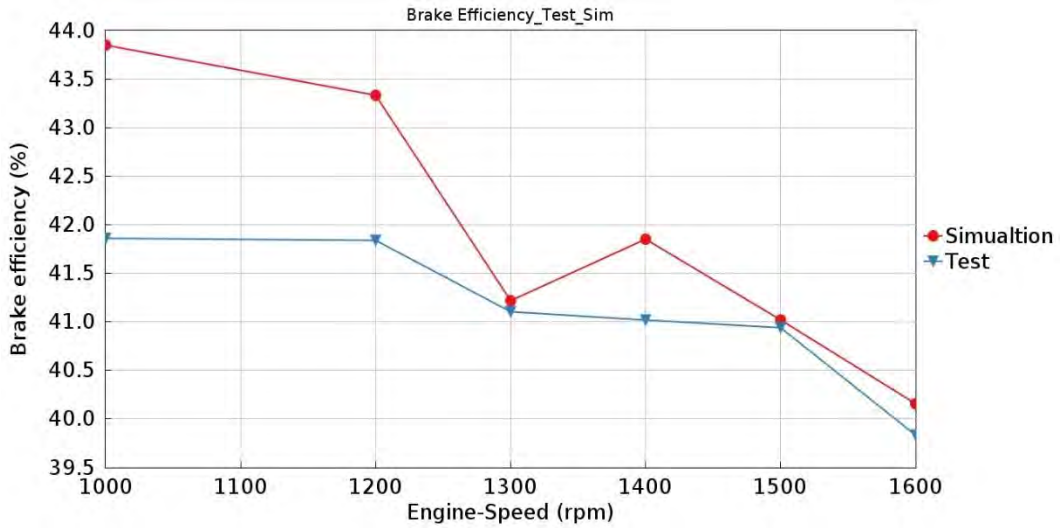

Figure 17: Brake efficiency from test and simulation

Figure 18 and Figure 19 show the high and lower pressure compressors efficiency maps respectively.

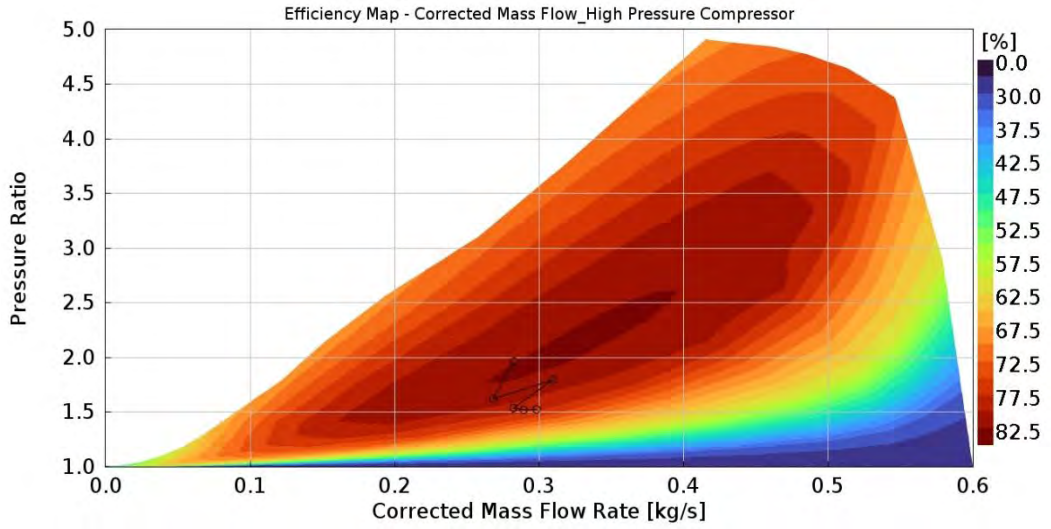

Figure 18: High pressure compressor efficiency map from simulation

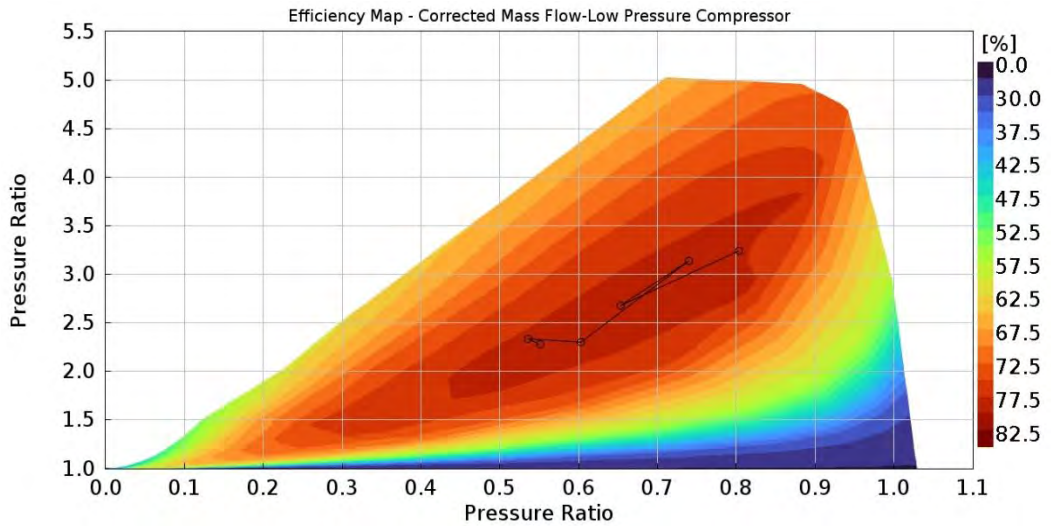

Figure 19: Low pressure compressor efficiency map from simulation

The results of simulation show that both compressors are working within good efficiency regions. Since the presented load points are from higher loads values, results of turbo system efficiency suggest that the size of turbo system for the current engine is good.

Conclusions from simulation results

By tuning the compression ratio, valve lash value and global heat transfer multiplier of the engine model based on test results, the model can estimate the engine performance with good accuracy. In the H_2 engine and for high load, high lambda condition, very high boost pressure and air mass flow is required. Therefore, some parts of the model (e.g. CAC system) requires more calibration for such high flow conditions. Otherwise, the VGT system can compensate in simulation and provide us with a performance that is not happening in reality.

Results from testing: Combustion performance

With nine months of testing, a huge amount of data has been collected. Below the methods of evaluation and some key results are presented.

Mixing and flame-propagation approach

In pre-mixed combustion, a sufficiently good mixing of hydrogen and air is required. Sufficiently good mixing means that a certain level of homogeneity has been achieved and that large scale vortices break down to small-scale turbulence during the compression stroke. With these conditions fulfilled at the time of ignition, stable ignition and flame-propagation can take place. Different approaches for good mixing are found in literature of light-duty gasoline DI SI engines. Such engine concepts are often based on port-induced air-tumble motion assisted mixing and vortex formation [11]. Swirl is another large-scale port-induced air-motion that has been shown to assist mixing [12]. The fuel (liquid gasoline) is in this case much heavier than the ambient cylinder gas, also in (partly) evaporated state. However, fever examples of mixing in HD LPDI H_2 ICE exist today. There is thus room for exploring different methods. Using hydrogen, there are several fuel properties which impact on the mixing in a different way compared to gasoline. Such parameters are density which for hydrogen is extremely low, diffusivity which is much higher and fuel energy content per mass which is much higher
as well.

The main research problem is thus to successfully mix a very light gaseous fuel with the much heavier ambient gas.

The momentum and global kinetic energy in the injected fuel is lowered by the low density and smaller needed amount of fuel mass but also compensated by the very high velocities produced at critical flow conditions of an under-expanded hydrogen. In total, the momentum of the injected fuel is lower than with a diesel-injection, but still significant.

In this work, Volvo designed the engine for a mixing-approach based on quiescent in-cylinder ambient air. Mixing is induced using the momentum of the injected fuel. To focus the momentum as much as possible, all fuel is concentrated to one single jet. The jet is further focused using an injector jet-guiding cap. The jet-guiding cap is protruding 9 mm below the firedeck. Mixing is also promoted by arranging for an as strong jet-piston interaction as possible. The strong interaction is created by directing the single jet towards the centre of the piston bowl. As a result of this geometrical setup, a toroidal vortex-system (Figure 20) is formed which develops in a symmetrical manner during the compression stroke [1].

The toroidal vortices act in a mainly axial-radial direction (like air-tumble) rather than tangentially (as with swirl). Since the ambient air is quiescent, the fuel can more easily mix by convection and diffusion and the complex turbulent toroidal vortices will interact and break down to small scale turbulence. During the compression stroke, the toroidal vortices develop while they gradually fill out the combustion chamber volume. Findings from internal pre-study activities with such a mixing concept in a smaller single-cylinder engine showed benefits by injecting the fuel as late as possible during the compression stroke. The latest possible end of injection (EOI) is determined by the requirement to achieve critical flow conditions. Critical flow occurs when the ratio between injection pressure and the cylinder-pressure is above two. The injection period is then phased towards the time interval when the piston speed is high (Figure 21).

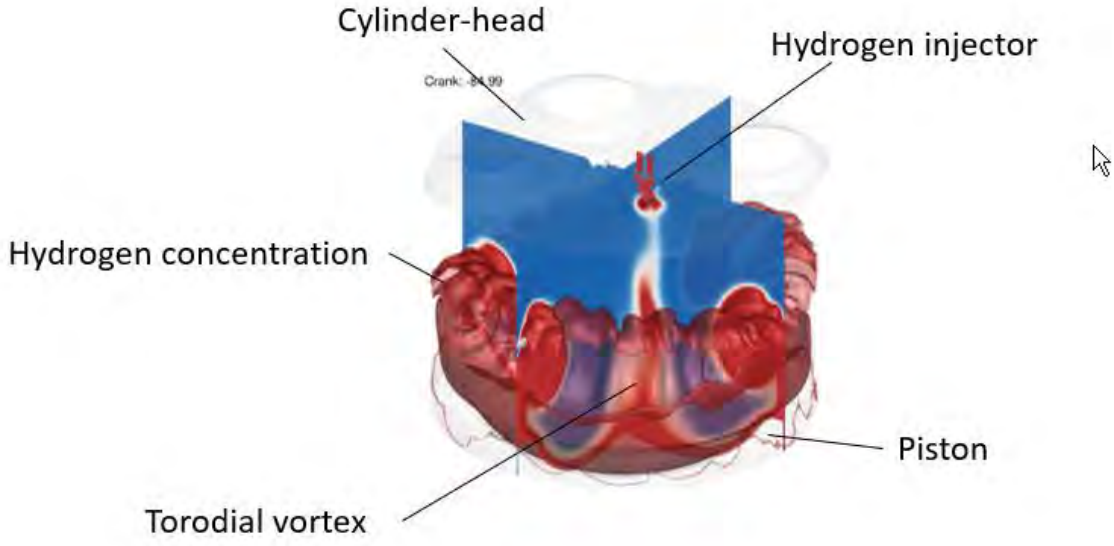

Figure 20: CFD Toroidal Vortex

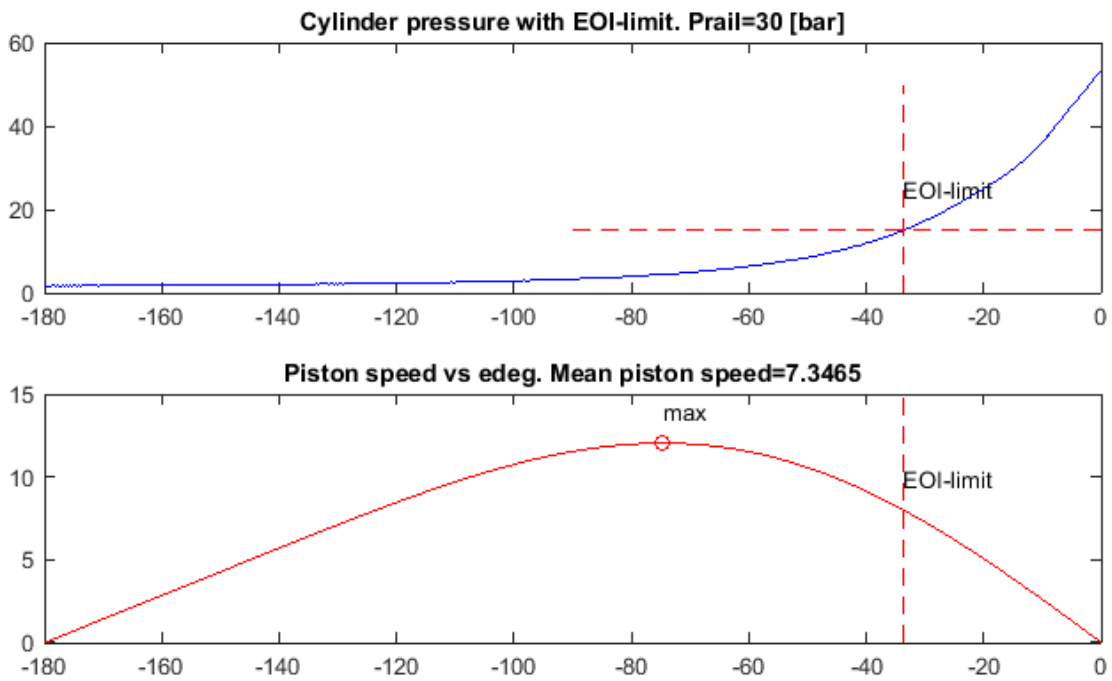

Figure 21: Piston velocity (example). Upper plot showing a limit in end of injection related to the fuel pressure/ cylinder pressure ratio required for critical flow. Lower plot shows piston velocity in m/s during the compression stroke.

This arrangement results in a larger relative velocity between the downwards going jet and the upwards going piston. The effect of jet-piston interaction (injection during compression) seemed to override the importance of long time for mixing (injection during intake).
Also supporting this approach is interesting findings in [13] which suggest that higher in-cylinder density during injection mitigates pre-ignition. Also supporting this approach is interesting findings in [13] which suggest that higher in-cylinder density during injection mitigates pre-ignition. At the time of ignition, it is important to avoid pre-ignition as well as misfire or partial burn. Stable and efficient ignition rely on multiple parameters such as the

level of homogeneity, level of turbulent kinetic energy and mean velocities in the near vicinity of the sparkplug itself. Stable and efficient ignition also relies on the physics of the spark [14]. During the mixing process, the ignition and flame-propagation is thus intended to develop in a symmetrical manner. The jet is ignited by a centrally located sparkplug allowing the flame to propagate radially with about the same radial distance to the walls in all directions.

Combustion system hardware

As described earlier in the section “Combustion system” (above) the in-cylinder air-motion can be considered as quiescent. A slight Miller-effect is achieved using a valve-system with inlet valve closure, IVC=545 at 0.2 mm lift. Main effect of interest is that closure of the inlet-valve thereby starts earlier which, under the condition to avoid injection during the air intake phase, allows to use an earlier SOI. Earlier SOI might be needed to reach the high load range (long injection duration). Injector and sparkplug are located in the region inside the inlet and outlet valves. However, none of these components is situated exactly in the very centre. The characteristics of the spark are controlled by a SEM capacitive ignition system [14]. A compression ratio CR=10.5:1 aluminum piston has a wide piston bowl with shallow sides. When the jet-bowl interactions start, the shallow sides lead the reflected flow upwards towards the liner in a favorable direction to promote the formation of the before mentioned toroidal vortex.

Evaluations methods

Evaluation is based on traditional averaged (slow) signals like speed, torque, H_2 supply-pressure (Prail), nitrogen oxides (NOx) emissions, air-flow etc. Fast-sampled signals were taken from each cylinder, mainly injector current, spark-plug primary current and cylinder-pressure (Pcyl). During part of the test-series, fast-sampled pressures in the inlet and exhaust manifolds as well as fast sampled Prail were available. Fuel injection phasing was defined by extracting start and end of injector energizing by analysis of the injector current profile (SOE and EOE). Spark-timing (ST) was determined from the primary current signal as first negative deviation from zero.

To enable comparisons with the combustion in other engines, gross indicated values were used (integration during compression and expansion strokes). By that, differences in gas-exchange and mechanical friction were omitted. In each cylinder, 200 cycles were sampled at 0.1 CAD rate. Using the pegged Pcyl curves, gross mean efficient pressure (IMEPg) and indicated gross efficiency (IEFFg) were calculated.

From IMEPg, the global stability parameters coefficient of variation (COV), lowest normalised value (LNV) were derived. LNV lower than 80% of the mean IMEPg was considered as partial burn and LNV<0 as misfire. Pre-ignition was indicated if a single Pcyl-curve at ST exceeded the mean Pcyl at ST with more than 5%. The Pcyl was also used to calculate maximum pressure gradient (dPcyl) and peak cylinder pressure (PCP). Finally, knock tendency was assessed using the parameter max amplitude of pressure oscillations (MAPO) here taken as the peak amplitude of the absolute values of a 4000 Hz highpass-filtered cylinder pressure-curve.

Testplans
The main objective was to explore drivability and performance over the speed-load range. Since the limits for stable operation for this LPDI prototype were not known before the tests started, an iterative procedure was needed to gradually establish needed conditions for stable operation. Tested engine settings were speed, injected fuel (load), lambda, injection pressure, injection phasing and spark timing. When stable operation was concluded at a certain load, the load was increased (longer fuel injection duration) initially with the same settings as in the previous and lower load-point but then gradually adjusted to mitigate any abnormal events or other limits occurring.

Engine test results
This section discusses the speed-load range, efficiency and abnormal combustion events. Emissions are discussed in a separate section below. As shown in Figure 22, tested operating points (OP) are mainly found within speed 1000-1600 rpm with maximum torque reaching 3000 Nm.

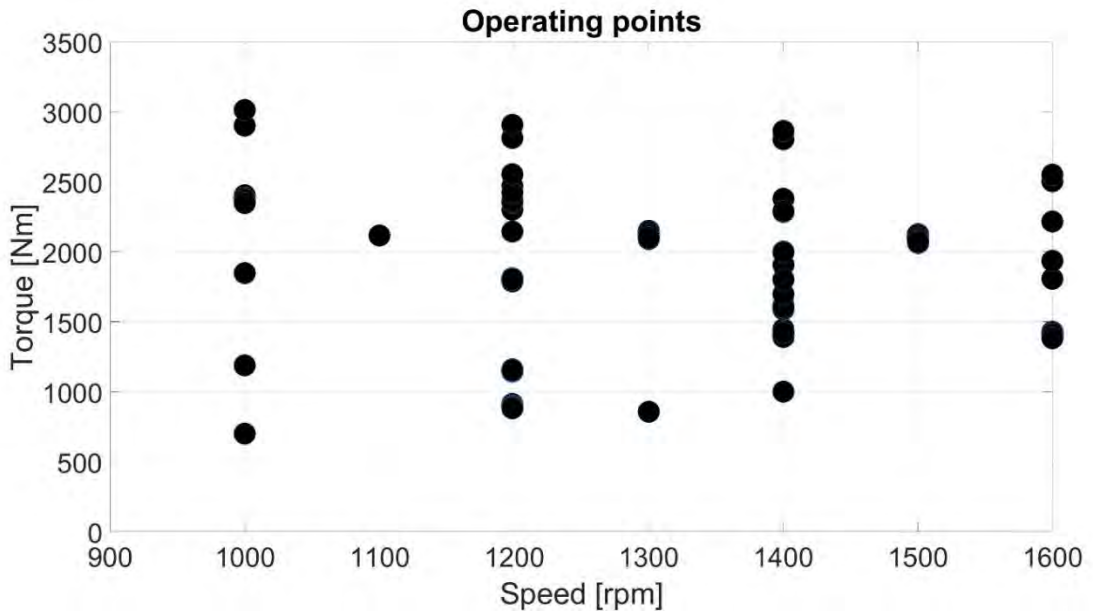

Figure 22: Overview of speed-load operating points (OP)

The basic performance of the engine is illustrated in Figure 23 where efficiencies versus variations of torque at different speeds are shown. Indicated gross efficiency (IEFFg), which only reflects the integrated work ∫pdV during the compression and expansion strokes, shows similar peak-values of 45% from low to about 80% load. In the highest load-range above 80% load,IEFFg slightly decreases. Brake efficiency, which also includes the gas-exchange and mechanical friction, shows 40-42% above 1000 Nm with decreasing brake efficiency when the load is decreased below 1000 Nm. Emissions of NOx were lower than 800 mg/kWh through-out these tests.

Statistics of abnormal combustion events were in each OP related to the total number of 1200 recorded Pcyl-curves. The major part of the measurements reached IMEPg COV<2% with IMEPg LNV > 90%. Generally, very few cases of misfires or partial-burn events were observed. At the highest load range, when the combustion was indeed stressed, some cases of pre-ignition occurred which lead to a sharp pressure gradient, high peak cylinder pressures and knock. Cases of late-cycle end-gas ignition knock occurring without any precedent pre-ignition were also registered.

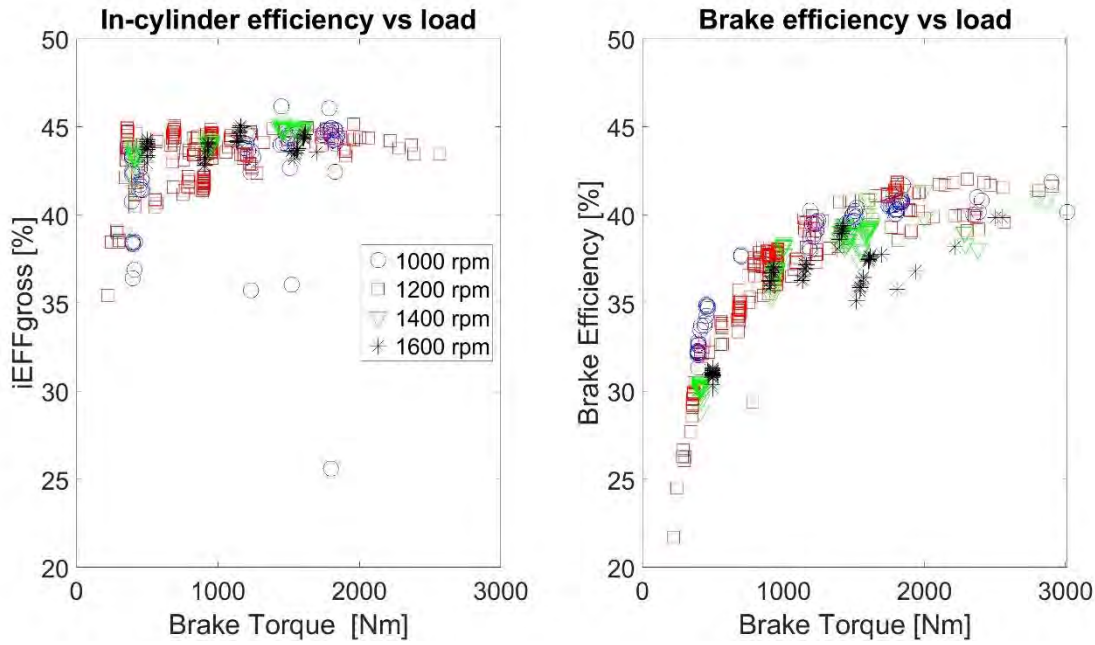

Figure 23: Load-swings screening

In common for these abnormal events is their stochastic nature. This is illustrated in Figure 24 which shows a series of cycles where the mid plot cycle 96 is a cycle for which knock is detected (MAPO=4.3 bar).

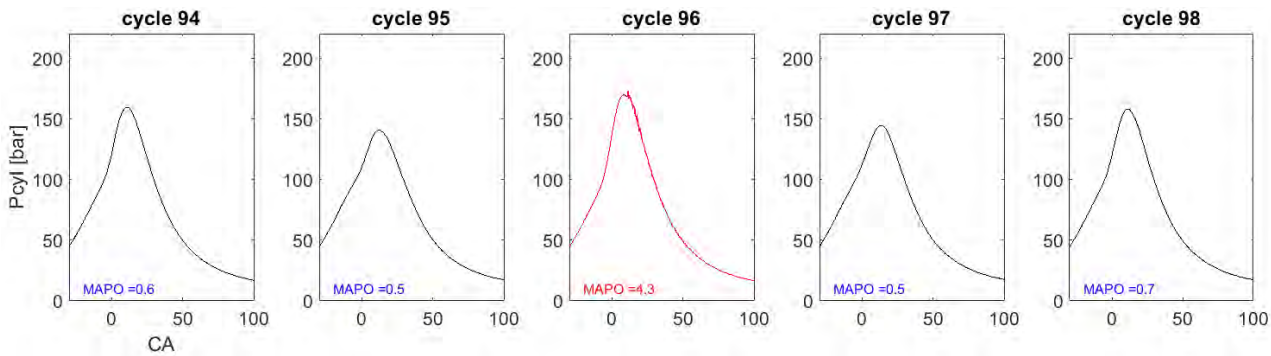

Figure 24: Typical series of cycles before, during and after a knocking combustion (mid plot)

Notable is that no relationship can be seen with the cycles just before, nor any effect on the cycles just after cycle 96. Stochastic pre-ignitions were attributed to hot-spot ignition due to leakage of carbon-containing particles. Stochastic low intensity late-cycle end-gas ignition knock events were attributed to variation in mixture homogeneity due to the turbulent nature of pre-mixed combustion. It can be observed that the knock-indication parameter MAPO is sensitive. Rather small fluctuations of Pcyl can render quite high values. It is therefore interesting to consider which levels of MAPO that should be deemed as critical. Following the example from [15], hard knock limit was defined as speed [rpm]/1000.

Discussion and conclusions (combustion)

Even though these results are based on an early prototype and limited to only steady-state operation, the results clearly demonstrate a good potential of an LPDI H_2 ICE which is based on a diesel engine with quiescent in-cylinder gas flow conditions. A rather high confidence-level for this conclusion is supported by previous studies using 1D-simulations, 3D

combustion CFD [1], experience from single-cylinder engine activities and literature which lead to the tested engine design. The combined features of the evaluated jet-bowl toroidal vortex mixing approach, single-jet based combustion system, dual-stage turbo-system and the SEM ignition system achieved up to 3000 Nm torque, up to 42% in brake efficiency at simultaneously low NOx-levels below 800 mg/kWh. At low to about 80% load, no significant abnormal combustion events were observed. At even higher load, some pre-ignition and knock events occurred, however at a low frequency.

Common for all operating points is that high levels of lambda>2.3 could be achieved. High lambda mitigates pre-ignition and knock and results in low NOx. Moreover, the favorable late injection strategy which takes place at higher in-cylinder density should also mitigate pre-ignition [13]. Common for all operating points is that high levels of lambda>2.3 could be achieved. High lambda mitigates pre-ignition and knock and results in low NOx. Moreover, the favorable late injection strategy which takes place at higher in-cylinder density should also mitigate pre-ignition. The observed frequency of abnormal combustion events at high load was thus low. Of special interest is that these knock events typically occurred as sudden stochastic single-cycle events rather than as a sequence of more and more severe abnormal events. Stochastic poor mixing quality due to the highly turbulent nature of pre-mixed hydrogen combustion could be one reason. However, secondary parameters which might create hotspots should be a more likely reason. Such secondary parameters could be stochastic existence of carbon-containing substance (lubricant, coolant) in the combustion chamber through leakage (valves sealings, injector and sparkplug sleeve, injector pintle to seat leakage), from reversed blow-by or caused by yet un-optimized piston ring-package function. Potential cylinder-internal improvement areas could be to reduce dead-volumes, smoothing of any sharp edges and improved cooling of hot components like the sparkplug and the exhaust-valves.

It is crucial to obtain stable combustion in all conditions, over time and including transients. However, it is a knowledge gap that knock acceptance criteria are not well established for HD LPDI H_2 ICE. To define acceptance criteria is thus an important research- and development area.

Results from testing: Engine out emissions

With the lean concept chosen, the potential to achieve very low engine out emissions (particularly NO_X) can be achieved. Below a summary of engine out emission results are described, including NO_X, particulates and greenhouse gas emissions (CO_2 and H_2)

NO_X emissions

This engine has a combustion system which seems to have very good mixing as was predicted and exemplified in Figure 26 where the 17L is compared to an SCE with a lateral injector position, the NO_X formation rate differs more than a factor two at higher lambdas between these two combustion concepts. The effect of the piston movement in the mixing process can be seen in Figure 29 where a minimum in NO_X formation is seen by injecting during the first half of the compression stroke. Earlier injection with more time for mixing does not give better mixing probably due to less interaction between piston and the gas jet. The double stage turbo gives enough air to achieve higher lambda than 2.4 in almost all the operation area which results in engine out emissions around the Euro VI levels or lower, Figure 25. There is possibilities to come down to NO_X levels below Euro VII by e.g. using the VGT to higher lambda or by later combustion, but it comes with loss of efficiency as seen in Figure 27 and Figure 28.With the 17L engine out NO_X levels the requirement of NO_X conversion to comply with Euro VII will be around 80% as with the Euro V Diesel, indicating that a EATS of EuroV specification (without oxidation catalyst) would suffice.

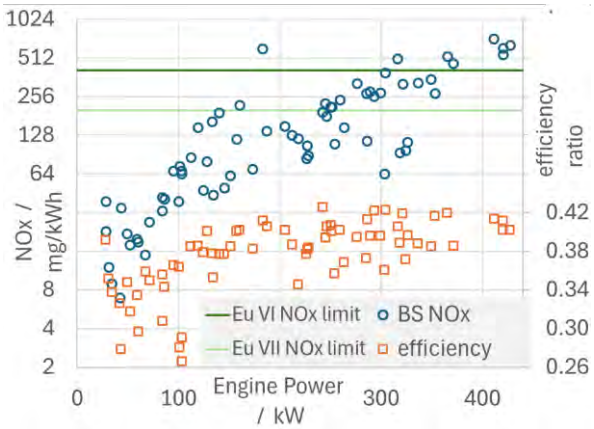

Figure 25: Brake specific NO_X emissions and Brake specific efficiency as function engine power for a wide range of speed and load

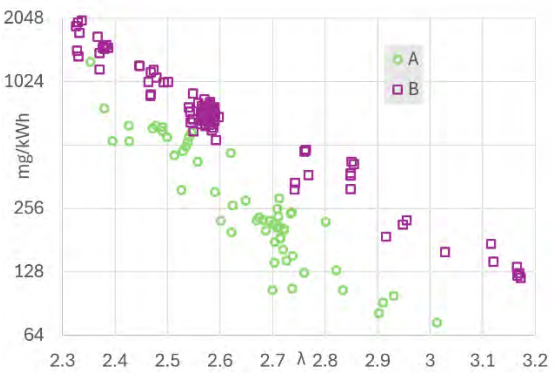

Figure 26: Engine out indicated specific NO_X for (A) the 17 L engine with central injector and for (B) a single cylinder engine with a lateral placed injector and fuel induced tumble IMEP 14 to 18 bar and 50% Mean Fraction Burned 5 to 14 crank angles after top dead center.

Figure 27: Brake specific NO_X emissions (o) and Brake specific efficiency (□) as function relative air fuel ratio (λ) at 1000 rpm and 1800 Nm and Mean Fraction Burned at 8° after Top Dead Center.

Figure 28: Brake specific NO_X emissions (o) and Brake specific efficiency (□) as function crank angel at 50% burned at 800 rpm and 1000 Nm and a relative air fuel ratio (λ) at 2.7.

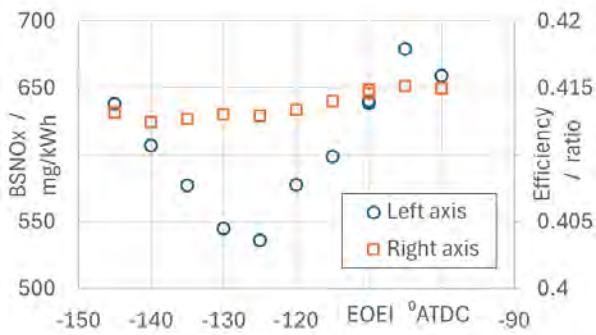

Figure 29: Brake specific NO_X emissions (o) and Brake specific efficiency (□) as function of End of injection at 1000 rpm and 1800 Nm and a relative air fuel ratio (λ) at 2.5, Mean fraction Burned at CAD 8 deg.

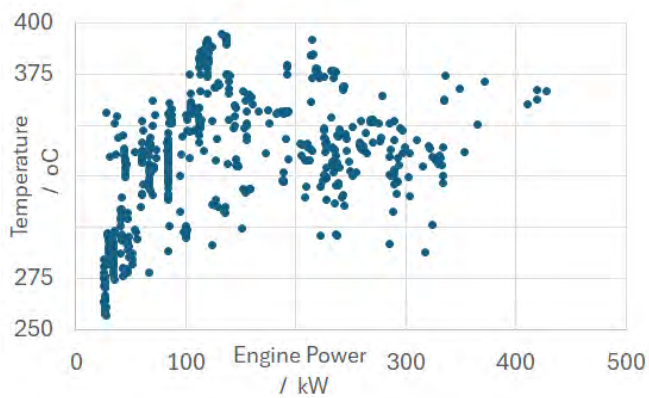

Figure 30: Steady state exhaust temperatures as function of engine power

Hydrogen emissions

The hydrogen emissions (Figure 31) are quite low and lower than a tenth of the legal requirement (40000 ppm in tailpipe). The hydrogen emission increases nearly linear with lambda. At lower power where lambda is high to avoid throttling the unburnt hydrogen in the exhaust can be in the range of 1-2 % of the fuel consumption (Figure 32). Unburnt hydrogen can also be found in the crankcase gases and are mainly below 0.5% of the fuel consumption. With a closed crankcase ventilation (CCV) this fuel can be circulated and used in combustion. The efficiencies reported in the figures above are with a CCV.

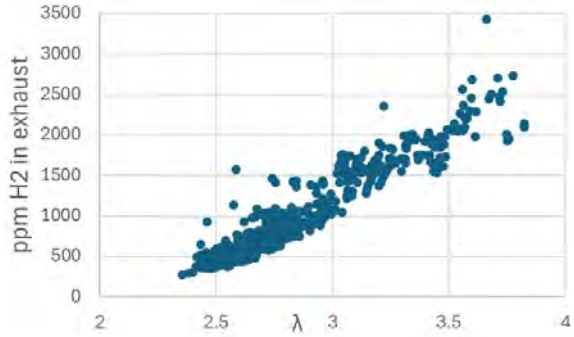

Figure 31: Hydrogen concentration in exhaust as function of relative air fuel ratio (λ) at various engine speed and torque, and ignition and injections timings

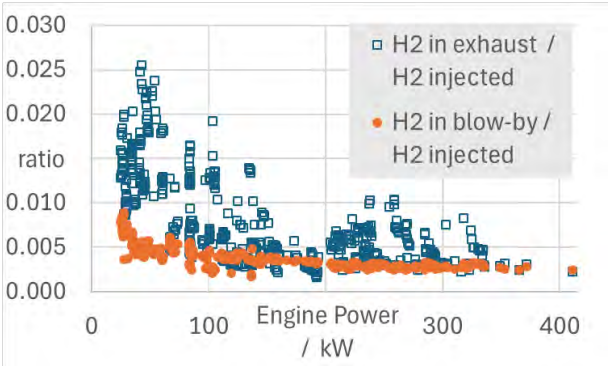

Figure 32: Relative fuel loss in exhaust and blowby as function of engine power.

Lubrication oil derived emissions

The level of carbon monoxide (CO) and hydrocarbons (HC) that are emitted from the lubrication oil consumption (Figure 33) are below the Eurov VII legislation (1500 and 80 mg/kWh for CO and HC respectively). The level of carbon dioxide (CO_2) is mainly below 0.5 g/kWh which was in the higher range from known oil consumption. Oil consumption can be measured more easily with the H_2 engine by measuring carbon content in the exhaust stream, but it can also be confused with coolant leakage into the cylinder. The PN23 (the number of particles larger than 23 nm) is around the range of the Euro VI legislation and hence the PN10 is higher than the Euro VII legislation which means that a particle filter is still needed. The main contribution to PN10 are particles in 15 nm size as seen in Figure 34Figure 34: Particle distribution at 1400 rpm and 14 bar IMEPg.. The PN10 is lower than has been reported elsewhere and one explanation can be is the avoidance of gas jet directed to the liner as shown by [16].

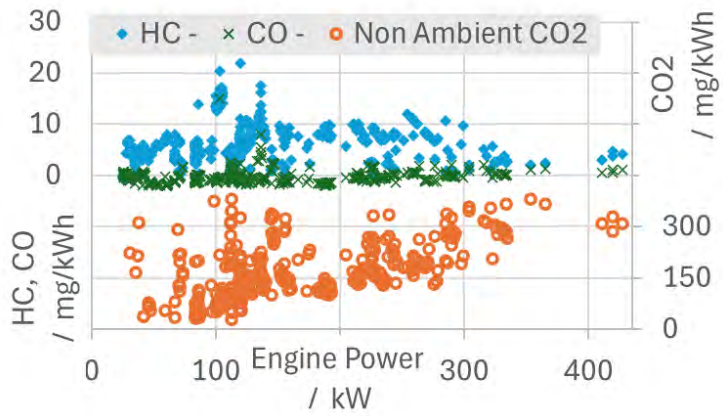

Figure 33: HC, CO and CO_2 emissions as function of engine power.

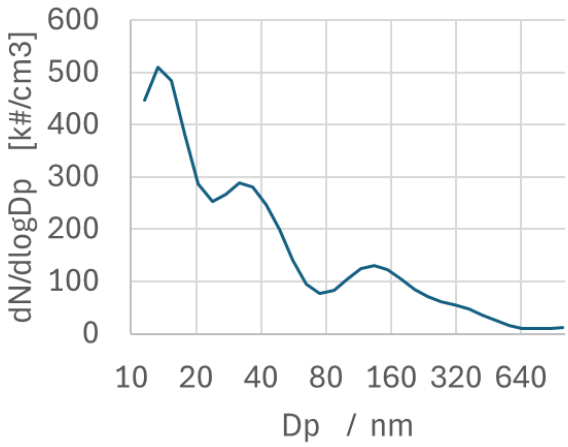

Figure 34: Particle distribution at 1400 rpm and 14 bar IMEPg.

Greenhouse gas emissions

Although HC, CO and soot are well below regulatory limits, an oxidation catalyst may be necessary to make the hydrogen engine a credible near-zero greenhouse gas emission alternative. Greenhouse gas emissions are very low compared to today's diesel, about 1% calculated as GWP100 (Global warming potential, 100 years average), presented in Figure 35. GWP20 can be above 5% of the GHG (Greenhouse gas) emission of today's diesel at cruising power (Figure 36). The main source for the engine out GHG emissions is the unburnt hydrogen in the exhaust, especially at low loads. Blowby gases can also be substantial GHG emissions source if an open crankcase ventilation is used. GWP from [17]

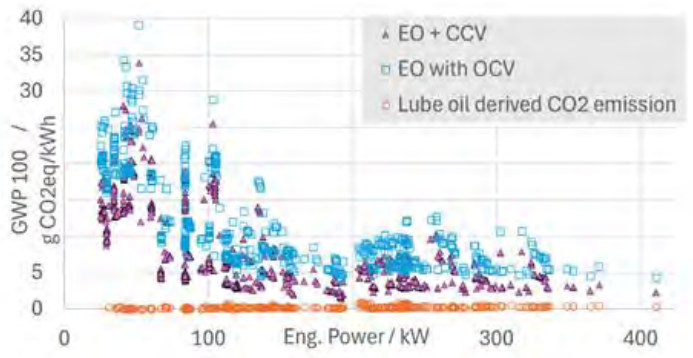

Figure 35: 100 years average of Global warming potential from engine without aftertreatment as function of engine power

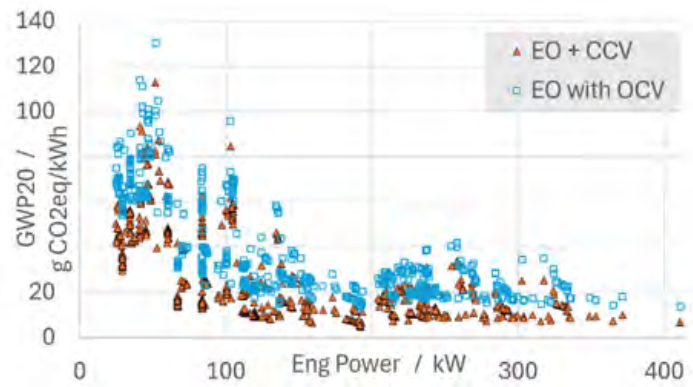

Figure 36: 20 years average of Global warming potential from engine without aftertreatment as function of engine power.

Conclusions from emission results

The engine out emissions are very low but still require an aftertreatment system. The aftertreatment for Euro VII can though be simpler than the Euro VI diesel with an SCR plus particulate filter. Since soot oxidation by NO_2 is not needed, the particulate filter can be placed after the SCR and catch particles from urea. Oxidation of hydrogen and unburnt lube oil can be performed by using ammonia slip catalyst designed for hydrogen oxidation also or by incorporate some oxidation catalyst in the particulate filter.

Volvo Ultra Lean

The definition that has been used for Volvo Ultra Lean in this paper means**:** a supported LPDI H_2 ICE engine with a pre-mixed, homogeneous, spark assisted combustion, limited with a minimum lambda of 2.5 in full load and part load as well as for transient condition.

Background

For a LPDI H_2 ICE it is common to aim for a lean combustion. The characteristics of hydrogen fuel means that both NO_X emissions and knock sensitivity will increase if lambda is decreasing. Because of these characteristics for hydrogen fuel, it is common to aim for lambda around 2.5. With a lambda of 2.5 the efficiency is high, and the level of knock, pre-ignition and NO_X emissions are low. An adaptive approach, to the characteristics of the hydrogen fuel, is to use Ultra Lean concept presented 2023 by Volvo in Vienna [1]. Therefore, the lower lambda limit is set to 2.5 and, if needed, one or more support systems are added to fulfil the requirements for performance, given this lambda min limitation. By doing this, we aim for high performance, high efficiency, increased robustness with almost no emissions.

The Ultra Lean system will increase the knock margin and improve the robustness.
Given a good fuel mixture formation, the NO_X emissions will be close to zero (Figure 37).

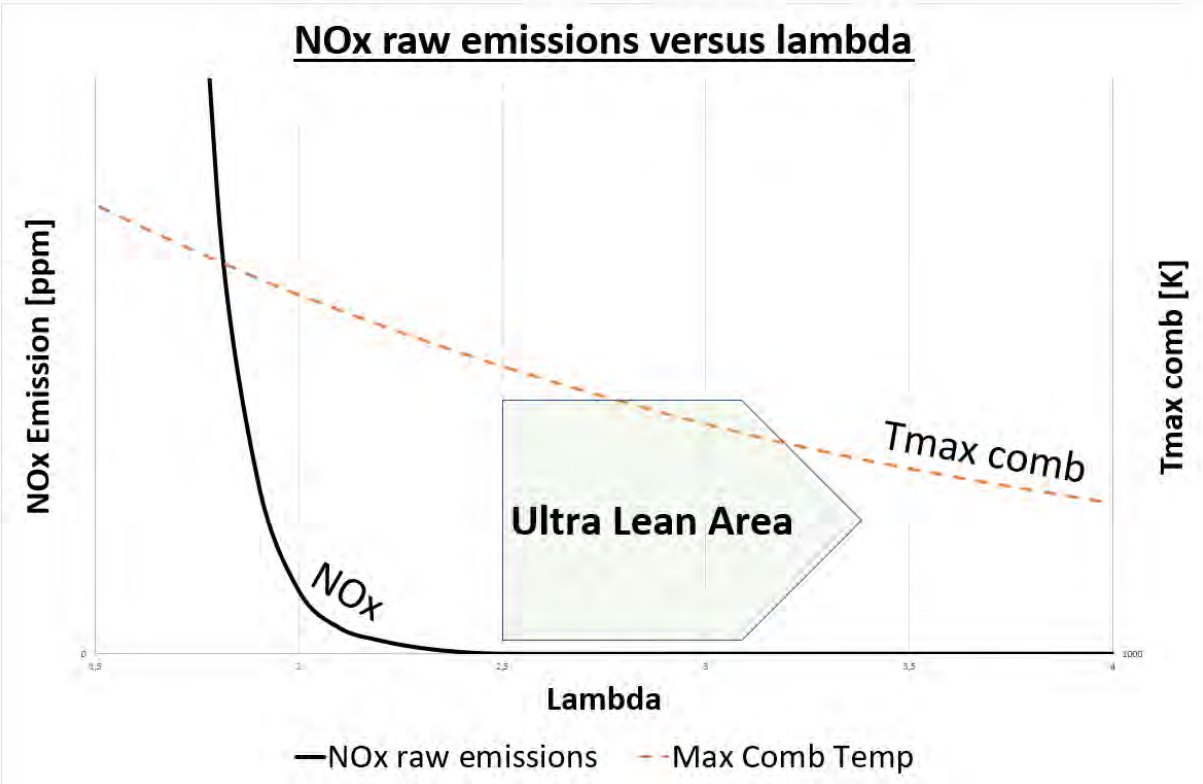

Figure 37: Schematic view of engine out NO_X emissions versus lambda

Experimental testing

The results from the multi- cylinder testing of the 17L dual stage engine are very promising showing both low emissions EO (engine out) in combination with a capability of Lean Combustion, i.e. Lambda 2.3 and above at lower torque. Thanks to a VGT as HP turbine with a fix geometry turbine as LP the setup was already present from the Volvo 16L engine, and an aerodynamic rematch was performed to suit the H_2 engine requirements. This has also been considered a viable solution for a lean H_2 ICE concept and was shown by Garrett [18]. However, even if this seems to be a good solution for steady state performance, but

transients, high altitude operation and warm climate can be hard to achieve without knock issues. The transient performance and boost margin in an air system may limit the capability to supress knock that can occur extremely rapid. Post injections, to improve the response from the turbo system, as described by Bosch in [19], can also be used if needed. Testing and development are ongoing.

Air Handling System

Volvo Trucks have a long history of sector divided turbochargers starting with the Volvo Viking in 1961 and continuing with the 13L EU6 Turbocompound engine [20]. The pulse utilization together with lower backpressures results in better scavenging due to the isolation of the 2 exhaust branches. For an H_2 engine, the improved scavenging (reduced residuals) is vital to avoid abnormal combustion. A SI engine, with its lower compression ratio compared to CI engines, also has a larger volume in cylinder after the exhaust stroke, resulting in higher residual fraction especially if the engine has a negative pressure ratio with high back pressure. To achieve minimum residuals with the help of scavenging a positive pressure ratio must be obtained across the cylinder. Thus, an engine with high pressure EGR is not suitable- rather using air for NO_X and knock control is more suited then. In addition, for an H_2 ICE with high lambda requirements, the exhaust backpressures will be high even with a reduced EATS enabled by the low EO NO_X and no HC and soot emissions. Thus, an engine with traditional boosting system consisting of turbos (single or dual stage) will operate with negative pressure ratios even with state-of-the-art components.

In the same manner how a turbocompound is suitable for an EGR engine concept [21] with the compound turbine contributing to EGR drive, an additional compressor on the intake side of an H_2 engine can give both the additional boosting needed, and also enable positive pressure ratio with good scavenging. Superchargers are traditionally either used for low engine speed boosting to improve transient performance or for increasing power in high performance engines over the full engine speed range. A traditional match was presented by Eaton in Aachen [22] for a hydrogen SI ICE. This showed the previously known effects of adding a blower to an ICE. In the same manner how a turbocompound is suitable for an EGR engine concept [21] with the compound turbine contributing to EGR drive, an additional compressor on the intake side of an H_2 engine can give both the additional boosting needed, and also enable positive pressure ratio with good scavenging.

Superchargers (SC) are traditionally either used for low engine speed boosting to improve transient performance or for increasing power in high performance engines over the full engine speed range. A traditional match was presented by Eaton in Aachen [22] for a hydrogen SI ICE. This showed the previously known effects of adding a blower to an ICE. However, the Volvo Ultra Lean concept-has the potential to solve all the abovementioned problems for a spark ignited H_2 ICE with large improvements compared to a traditional setup while minimizing the negative sides of mechanical boosting. The Ultra Lean engine concept setup is schematically shown below (Figure 38).

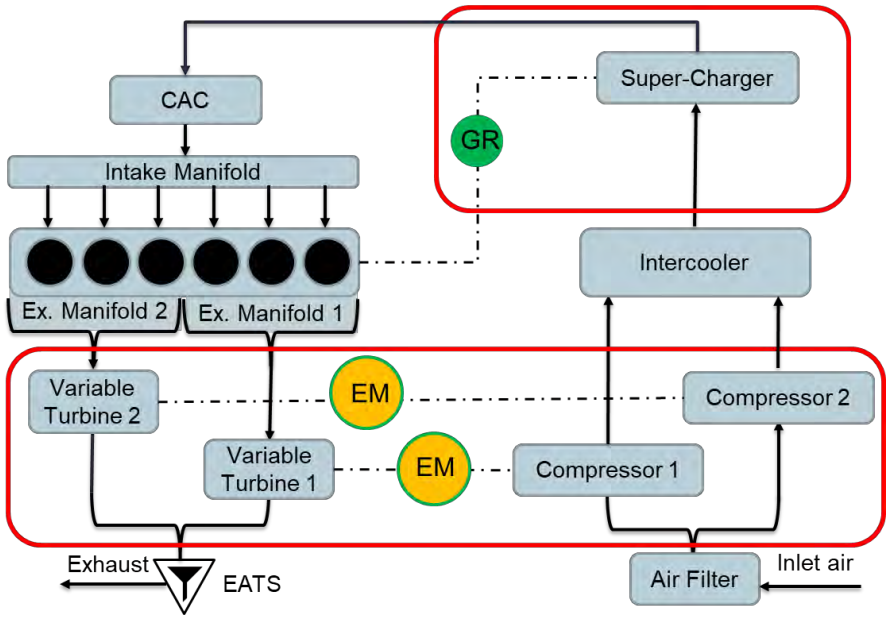

Figure 38: The H_2 Ultra Lean engine schematic

Simulations of a 13L LPDI H_2 ICE Ultra Lean concept

Further evaluations have been performed in GT-suite simulations for the concept that consist of a dual stage boosting solution with parallel dual LP e-turbos followed by an HP compressor driven by the crankshaft. Here the result is positive pressure ratio over the engine with high boost levels, high Lambda and a secured scavenging in all operating points. In this stage, a 12.8 Liter engine simulation model for an LPDI concept was used as base model. A camshaft with similar characteristics as the one used in the 17L engine test was used, i.e. with an earlier intake valve closure (slight Miller, IVC 549). Compression Ratio was increased in the second simulation round. The heat release rate model was based on a Wiebe function matched to single cylinder H_2 ICE test data for a wide range of lambda, IMEP and engine speeds, all without EGR. Simulations were performed with combustion CA50 @ 8deg after top dead centre (Atdc). A knock model was also used to monitor eventual knock tendency.

The simulations performed are rather exhaustive to explore the potential fully. Therefore, a meta model was tailored to the results to get part load mapped data as input for complete vehicle simulations. 8 full load and 21 part load points, a total of 29 operating points between 800-1800rpm and up to 21 bar in BMEP were simulated (Figure 39).

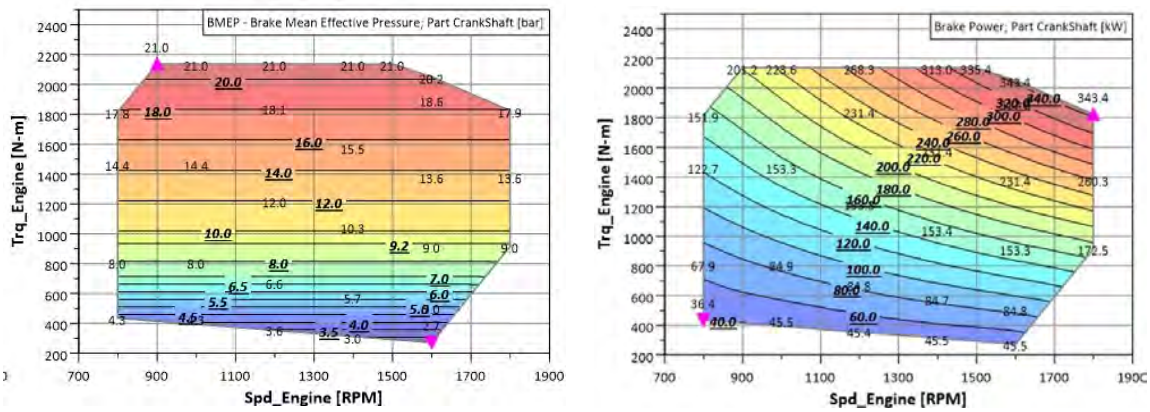

Figure 39: The 29 operating points that were simulated, and engine torque curve with power

In the simulations sweeps of several air system actuators were performed resulting in 2300 cases/ point: Turbo-speed, VGT rack positions and SC gear ratio (GR); represents a SC with variable GR, and e-turbos with capability to add energy for boost or to generate electrical compound power.
Post processing of the data enables the study of the effect of different electric motor (EM) power levels (limitations) for the e-turbos and different strategies for SC gearing, for example fixed GR or even bypass solutions.
The concept enables very efficient scavenging with high lambda for the combustion, separated exhaust branches with single entry bi-turbos, thus minimized back pressure, with no detrimental effect in turbine efficiency at unequal admission (as for a single pulse divided turbine). The e-turbos enable a harvesting of the energy which is vital since the engine has a very high flow at Ultra Lean conditions. By this, energy from the crankshaft that is used for boosting can be harvested back in the e-turbos after it has supported the combustion at Ultra Lean conditions.

Turbo compressor (Figure 42) and variable geometry turbine with state-of-the-art efficiency were chosen from other Volvo heavy duty engine projects.

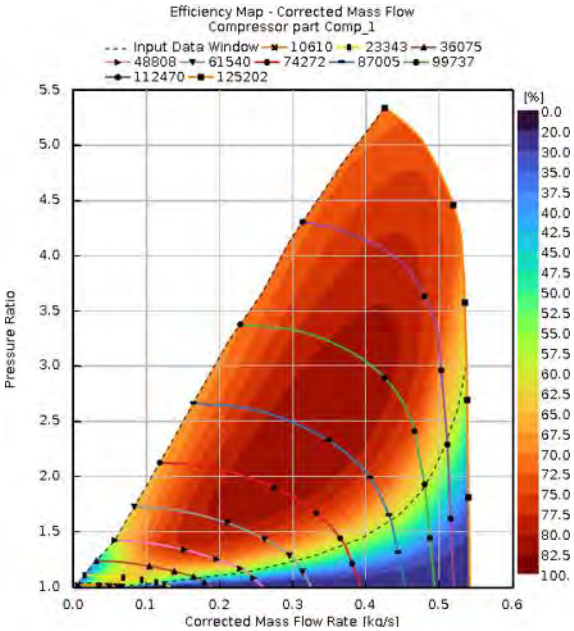

Figure 41: LP Turbo Compressor map (used x2)

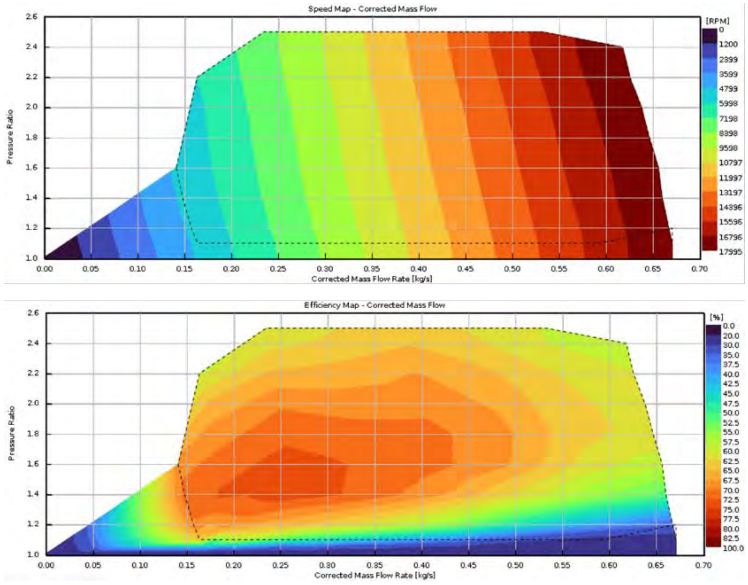

Figure 40: 1900cc HP Roots blower map

For the HP compressor a roots type supercharger was chosen (Eaton R1900) since a displacement compressor has several benefits in such an application (Figure 41).
Matching the supercharger displacement and gearing to the piston engine results in an additional compression before the airflow is entering the inlet manifold. The linear behaviour in turn enables having a fixed GR and constant engagement for the HP SC if a variable LP stage is utilized (VGT in combination with EM). Higher GR result in higher pressure ratio in the roots supercharger, which for the LP turbo compressor mimics a larger displacement engine.

Simulation results

With the added energy flow paths in the system, efficiencies must be defined:

- Brake efficiency is the efficiency that comes from the crank shaft power. Therefore, it always includes the SC power (Cost to drive the SC), pumping power and friction power
- Net efficiency is defined as the efficiency that comes from the crank shaft power + the E-shaft power (electrical power from e-biturbo shafts)
- Positive E-shaft power (produces power), Net efficiency > Brake efficiency
- Negative E-shaft power (consumes power), Net efficiency < Brake efficiency

The results shown below are optimized for best Net efficiency. A flat Net efficiency trend is achieved (Figure 42) thanks to the compound energy from the e-turbos (E-shaft power). Lambda is also flat relative to engine speed (Figure 43), enabling Ultra Lean conditions at all loads. Up to 1100-1200 rpm the pumping power is positive (Figure 43).

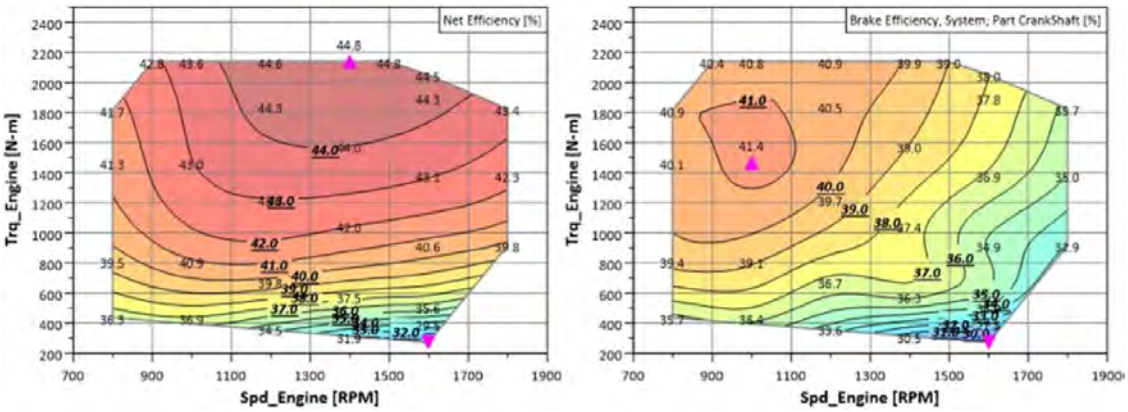

Figure 42: Net efficiency, left and Brake efficiency, right

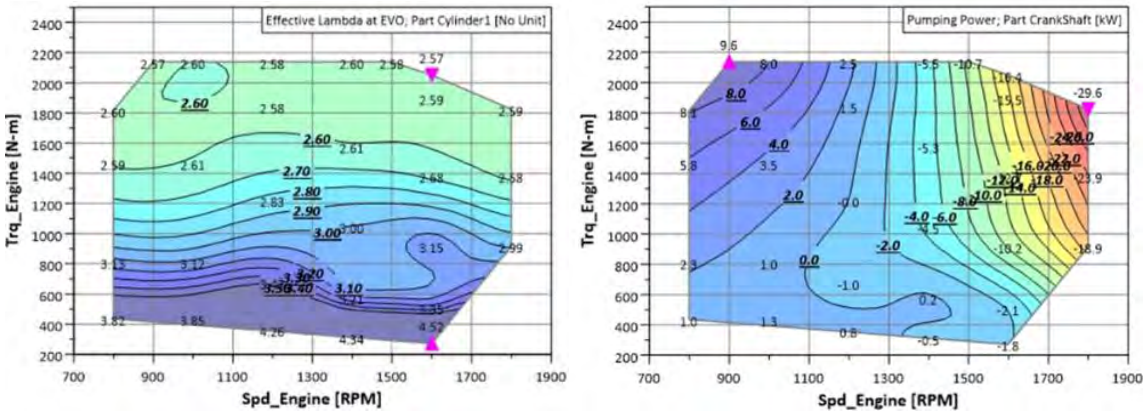

Figure 43: Lambda, left and Pumping power, right

The power diagrams (Figure 44, Figure 45) show that the compound power compensates for the supercharger attachment power at higher engine speeds and loads which is the reason for the flat net efficiency vs. engine speed.

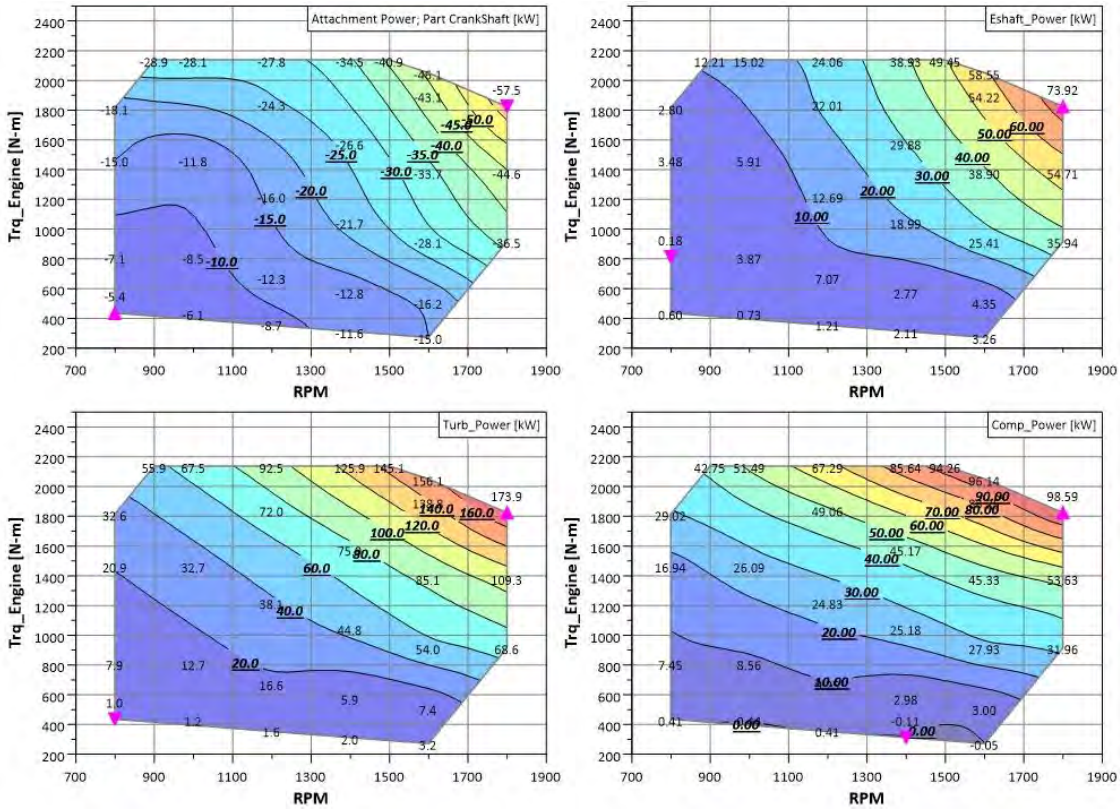

Figure 44: Clockwise from top left: HP blower attachment power, E-shaft power, LP compressor power and turbine power

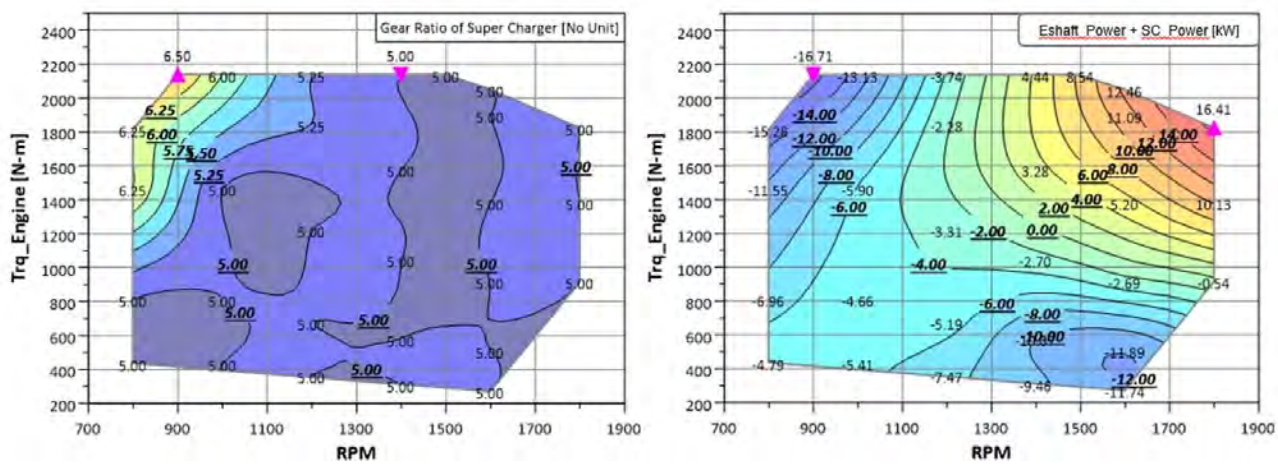

Figure 45: E-shaft power + SC (attachment) power, left, SC (attachment) gear ratio, right

With the dual compression stages, the engine receives a positive pressure ratio, thus a positive gas exchange loop is realized at lower engine speeds. In the PV-diagram (Figure 46) at 1000rpm full load the inlet manifold pressure is 3.8 bar and the exhaust manifold pressure is 2.8 bar. This results in positive net work also in the gas exchange loop. The gas exchange work is usually negative (without the HP compressor stage). The SOI is just at IVC close to bottom dead centre.

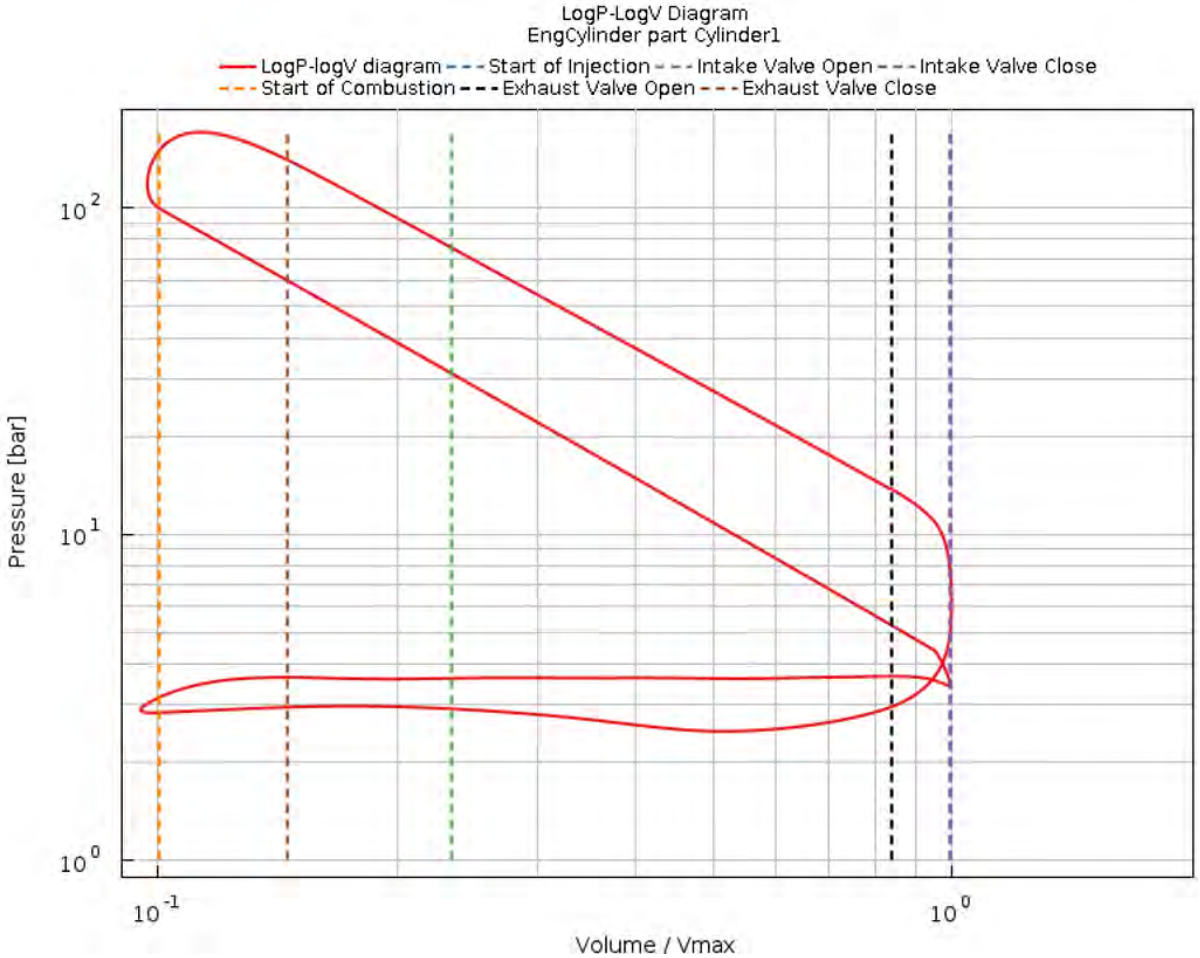

Figure 46: PV diagram for a low-speed high load point: 1000rpm, 21Bar BMEP, GR 5.75

In the figures below (Figure 47, Figure 48) operating points for best Net efficiency are plotted in the LP compressor and HP supercharger maps.

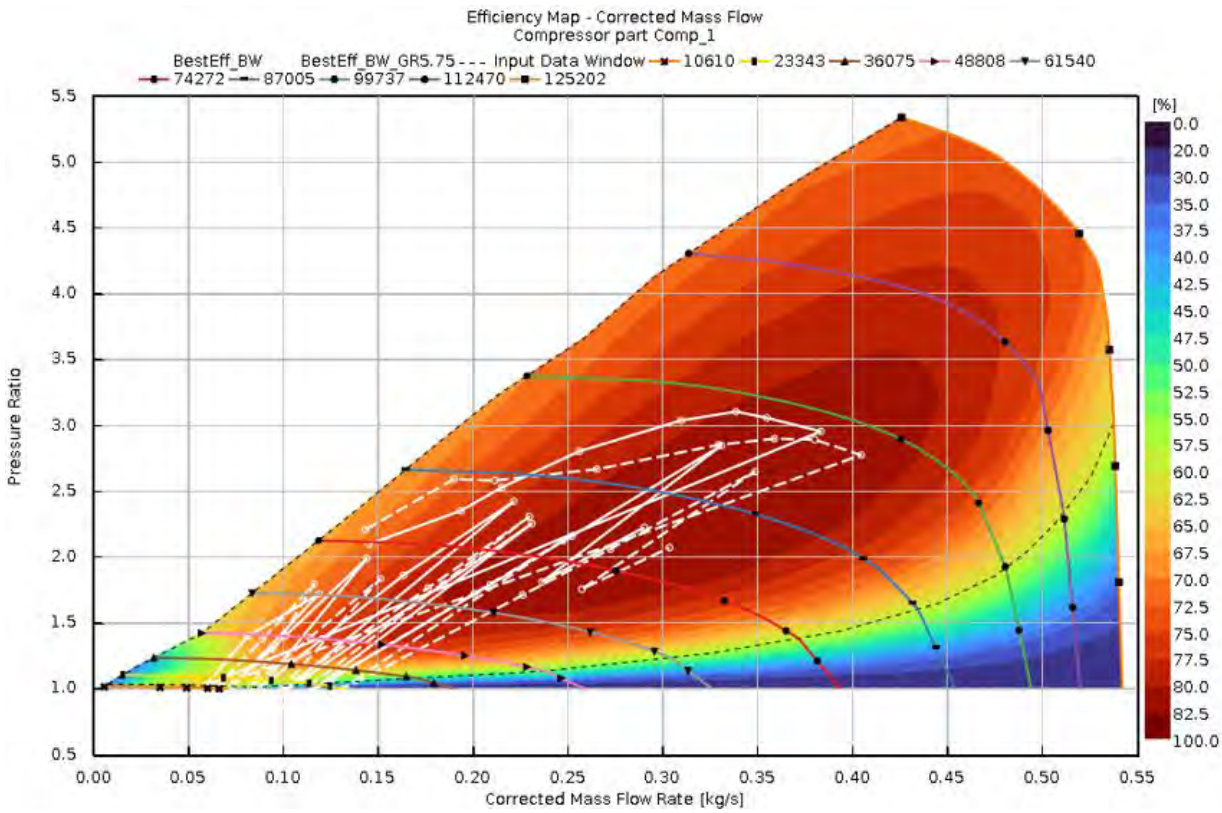

Figure 47: Compressor map for the LP stage, Best efficiency (solid line) and fixed GR 5.75 (dashed line)

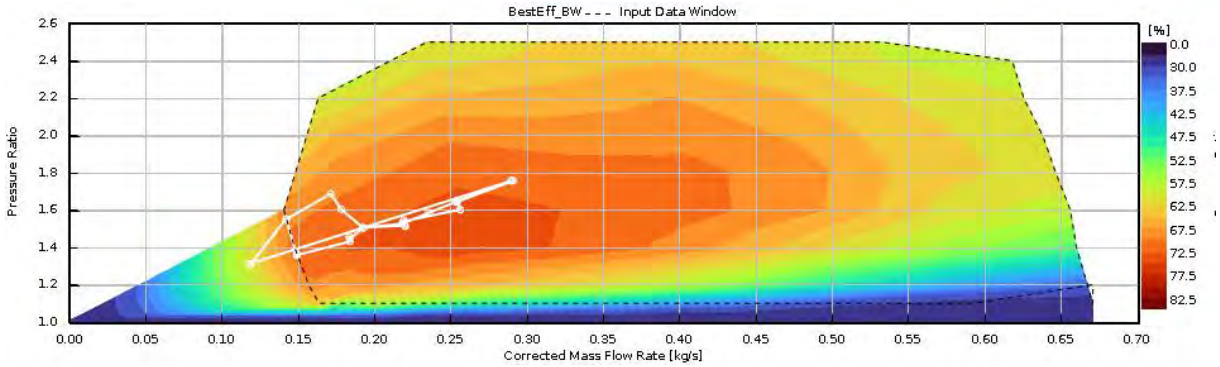

Figure 48: Compressor map for the HP stage supercharger, Best Efficiency (solid line), variable GR

To maximize engine system efficiency, the VGT is operating at low rack positions to extract energy that is both driving the LP compressors and generates electricity in the Electric Motors (generating) on the turbo shafts. For an optimization towards ~0 kW E-shaft power, the net efficiency drops between ~1-4% points (equal to brake efficiency without any E- shaft power). The efficiency drops most at high speeds and high load. This reflects a setup where the Ultra Lean concept is without any e-turbos, instead using free- floating VGT turbos. The E-shaft power can either be used directly in the vehicle or stored. If it is used directly, the combined torque from the engine system (driveline) would be according to the left plot in Figure 49, with a system combined peak power of 417 kW (343kW brake power) at 1800rpm. 74kW of this system combined peak power is electrical power.

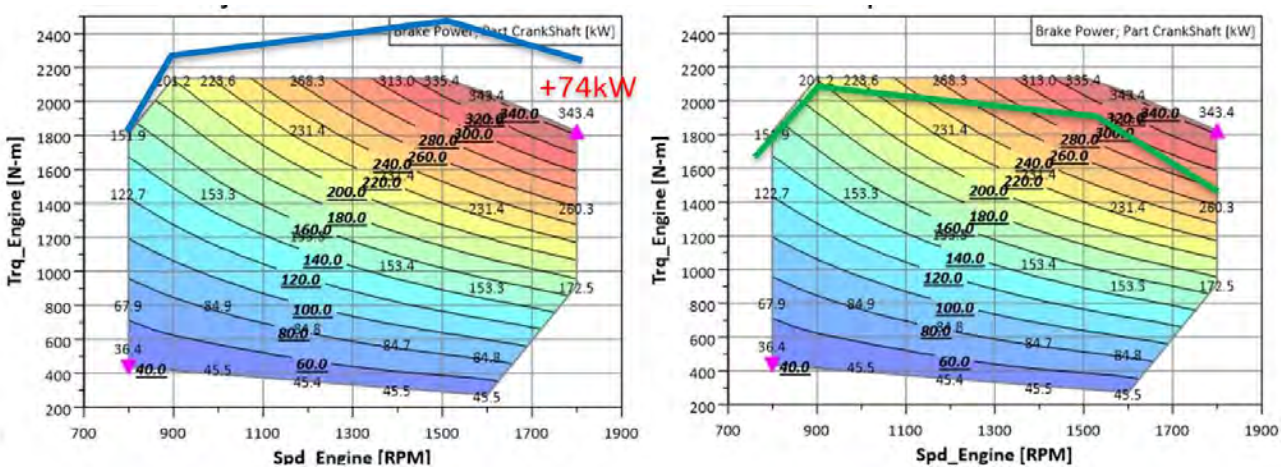

Figure 49: System combined torque curve, left. System reduced torque curve, right.

Another option is to reduce the ICE brake torque curve with the E-shaft power according to the right in Figure 49. This enables a further optimization of the engine at lower power levels. Note that the system combined torque curve is still according to the classic Volvo 460hp torque curve with constant torque line followed by a constant power line. This system has peak power of 343kW like the original brake power (Volvo 460hp engine) but with a reduced ICE brake power to about 300kW at 1500-1600 rpm. The E-shaft power fill in the gap from the reduced torque curve up to the classic Volvo 460hp torque curve.

Air handling summary and conclusions

17L LPDI engine tests show that high lambda and very low emissions with stable combustion is possible with a suitably matched Dual Stage Turbo system. VGT adds flexibility and improves the width for the system.

Recent simulations of the Volvo LPDI H_2 ICE Ultra Lean concept on 12.8L engine show results with high engine efficiency over a wide speed range with a max combined Net efficiency close to 45%. With good scavenging and high lambda, Volvo Ultra Lean concept is robust with very low emissions engine out in combination with good margin to combustion instabilities and excellent transient potential. The Volvo Ultra Lean concept can be tailored with a further improved engine efficiency matched for lower BMEP with the help of the E-shaft power for torque fill-in (LP turbo compounding), optimizing camshaft and CR. Furthermore, the concept can be tailored to reflect a 560hp engine by using the E-shaft power in a hybrid solution together with the ICE crankshaft torque according to the Volvo 460hp 21bar BMEP torque curve. The Volvo Ultra Lean concept can further be optimized regarding system sizing and friction losses depending on the chosen application.

H_2 ICE Ultra Lean with hybridisation

There are several alternative ways of implementing an Ultra Lean concept by using various levels of electrification and supported boosting systems. A modular alternative for an Ultra Lean LPDI H_2 ICE engine can be an integration with a PHEV system (Plug-In Hybrid Electric Vehicle). The P2, P2.5 or the P3 (Figure 50) systems are attractive combinations from a modular and a complete system perspective. One important feature is the pure electrical drive possibilities but also efficiency, performance, driveability, range and emissions will be competitive when combined with a LPDI H_2 ICE designed to be a part of a hybrid system.

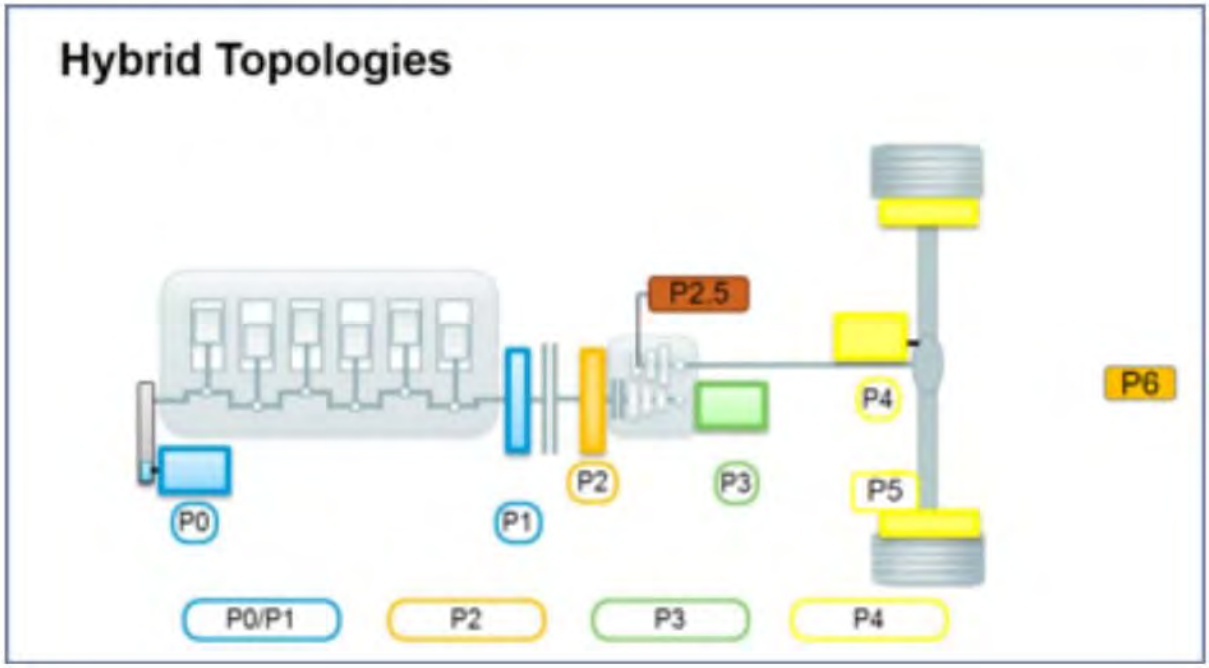

Figure 50: Hybrid Topologies

The Ultra Lean LPDI H_2 ICE is limited with a very lean lambda during transient operation. The transient torque response from the engine will then be delayed by the dynamics of the turbo system. In an Ultra Lean hybrid system, the electrical motor continuously supports the LPDI H_2 ICE engine by adding the torque deviation between the requested torque and the actual crankshaft torque from the engine. The total output torque, or the sum of the torque from the engine and the hybrid system, will then follow the requested torque. (Figure 51).
As a result, the lambda can be kept above 2.5. The system torque response will follow the requested torque without delays, while NO_X emissions are close to zero. For negative transients, the use of throttling can be avoided. Since throttling can be avoided, the positive differential pressure over the piston rings will reduce both hydrocarbon and particles emissions originating from oil residuals. Instead of using the throttle, the negative torque ramp can be controlled with the use of negative torque from the hybrid system. Therefore, the system torque request can be fulfilled without the need for throttling. With a sophisticated hybrid torque coordinator, the hybrid system can support the LPDI H_2 ICE with torque assist, both for positive and negative transients as described. Therefore, a substantial part of the emission control can be performed by the hybrid system since most of the engine out emissions can be avoided before they are produced.

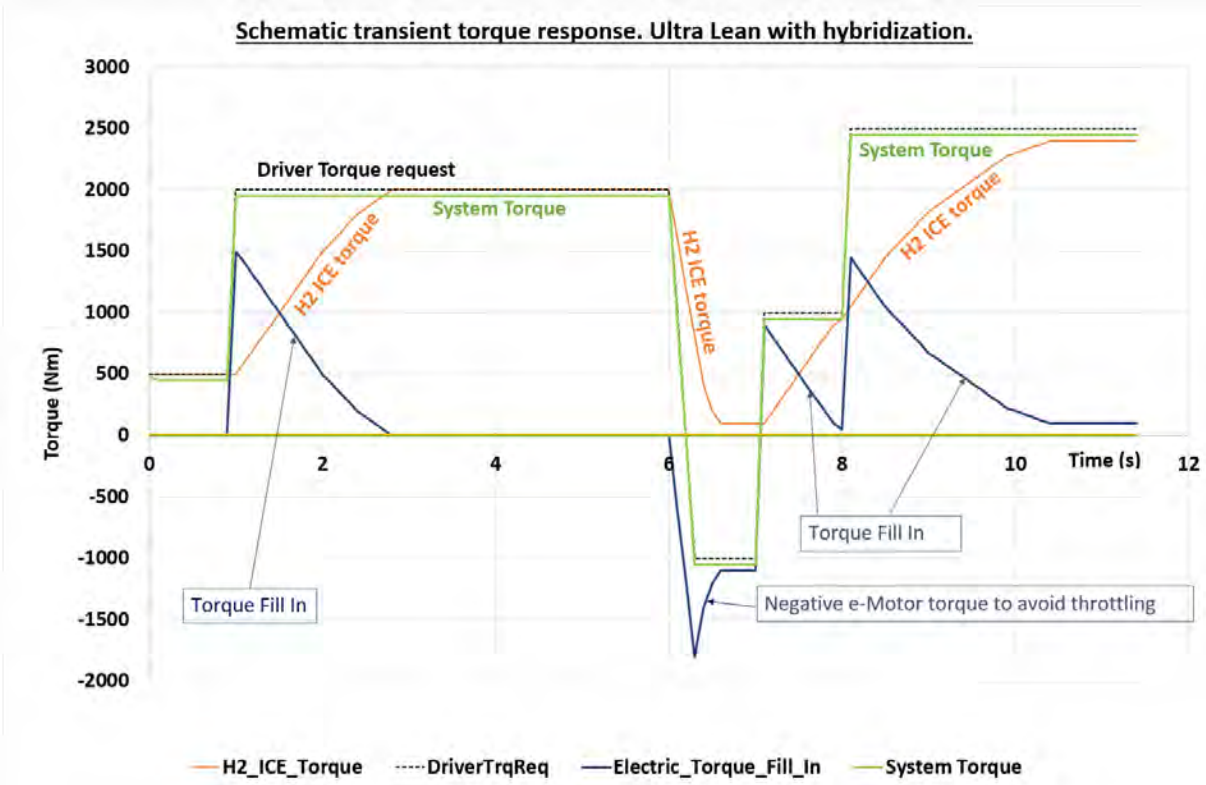

Figure 51: Schematic view of torque coordination for a hybrid system

The hybrid system torque control strategy must be co-ordinated with a developed battery SOC (State of Charge) strategy. Therefore, the torque assist must always be available to actively support the engine with torque control for:

- Ultra Lean for low emissions
- Driveability
- Optimized efficiency
- Regenerative braking
- Engine brake performance
- Escape and limp home functions

When battery SOC level is high enough, there should also be support for:

- Pure electric drive
- Start/stop
- Torque and power boost “on demand” with combined ICE and Hybrid torque

Thanks to the torque assist from the hybrid system, the design requirements for the LPDI H_2 ICE engine can be re-balanced. For robustness, and consistent power delivery also with a low battery SOC level, the LPDI H_2 ICE in a hybrid installation should be designed for high power output. Turbo matching and turbo system design should be optimized for high airflow, high boost pressure and high efficiency but with less requirements for time-to-torque. Therefore, a large and capable two stage turbo system can be used. Since the hybrid system can compensate for the dynamic drawbacks of Miller cam timing, the compression ratio can also be increased to improve the efficiency.

Conclusions for Ultra Lean with hybridisation:
Combining a LPDI H_2 ICE Ultra Lean engine with a hybrid system is a very good match. The unique possibility with the very lean pre-mixed spark ignited hydrogen combustion, producing a very low amount of engine out NO_X emissions, can be fully explored without the drawbacks of slow torque response and low performance.

In a hybrid system, the LPDI H_2 ICE can be optimized for high power and high efficiency by using a 2-stage boosting system combined with Miller cam timing and increased compression ratio.

The hybrid system will contribute to features like increased range, improved engine brake performance and pure electric drive including escape and limp home function. Combining the power from the hybrid system and the combustion engine, also for static use, will increase the complete system performance to be very competitive.

Volvo 17L LPDI: Summary and outlook

This paper shows that a spark ignited hydrogen internal combustion engine with low pressure direct injection can both be powerful, fuel efficient and release low nitrogen oxide emissions. Keys to this achievement are the optimised combustion system (with centrally placed fuel injector and spark plug) and a powerful boosting system that is able to keep in-cylinder lambda at very lean levels (lambda >2.5) during steady state operation.

To ensure that the same lambda is kept during transient operation the application of an electrified boosting system could be applied. While increasing system complexity and cost it could improve driveability while always keeping emissions at very low levels.

Even though the engine is successful but obviously there are still improvements to be found, like even better mixing to further supress risk of pre-ignition.

On top of that many factors are still to be defined, like acceptance levels for knock and other combustions stability parameters.

Abbreviations

1D, 3D	1D-, 3D-dimensional
atdc	After top dead centre
Battery SOC	Battery state of charge
BMEP	Brake mean efficient pressure
Brake efficiency	Efficiency based on the power from crankshaft
CA / °	Crank angle, here -360° to 360° where TDC combustion=0°
CAC	Charge air cooler
CA1090	Duration from 90 to 10 heat released
CA50	CA at 50% fuel
CCV	Crank case ventilation
CFD	Computational fluid dynamics
CR	Compression ratio
DI	Direct injection
DME	Dimethyl ether
DST	Dual stage turbo
DUR	Duration (injection, combustion)
EM	Electric machine
EOE	End of energizing. Based on injector current signal.
EOI	Actual end of injection
E- shaft power	Electrical power from e-Turbo shafts
GHG	Greenhouse gas
GR	Gear ratio
Gross	Compression and expansion strokes -180° to +180°
GWP	Global warming potential
H_2 ICE	Hydrogen internal combustion engine
HP	High pressure
HPDI	High pressure direct injection
HVO	Hydrogenated vegetable oil
ICE	Internal combustion engine
IMEP	Indicated mean effective pressure
IVC	Intake valve closure
BMEP	Brake mean effective pressure
PMEP	Pumping mean effective pressure
Lambda	Lambda based on fuel flow and fresh air-flow measurement
Lambda_exh	Lambda based on exhaust composition
LNG	Liquid natural gas
LP	Low pressure (fuel injection), Low pressure (turbine)
LPDI	Low pressure direct injection
Net	Complete 4-stroke cycle -360 to 360°
Net efficiency	Efficiency based on brake power+ E- shaft power
NOx	Nitrogen oxides
OP	Operation point
PHEV	Plug-In Hybrid Electric Vehicle
PN	Particle number

SBTi	Science based target initiative
SC	Super charger
SI	Spark ignited
SOC	Start of combustion (detected from in-cyl indication)
SOE	Start of energizing. Based on injector current signal.
SOI	Actual start of injection
ST	Spark timing
TDC	Top dead centre
UL	Ultra lean
VCB	Volvo compression brake
VGT	Variable geometry turbine
WHTC	World harmonized transient cycle

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